

Now, the differential star count formula is, from Eq. (24.4),

$$\begin{aligned}
 A_M(M, S, \Omega, m) &= \frac{d\bar{N}_M(M, S, \Omega, m)}{dm} \\
 &= \frac{\ln 10}{5} \Omega n_M(M, S) 10^{3(m-M+5)/5} \\
 &= \frac{3 \ln 10}{5} \bar{N}_M(M, S, \Omega, m). \tag{24.5}
 \end{aligned}$$

If either $\bar{N}_M(M, S, \Omega, m)$ or $A_M(M, S, \Omega, m)$ is known from observations, the equations can be used to determine the spatial number density $n_M(M, S) dM$.

The constant-density model just described suffers from a dramatic flaw. When it is used to calculate the amount of light received at Earth due to the stars contained in the solid angle Ω , the result diverges exponentially as m increases (Problem 24.4). This implies an infinite amount of light arriving from infinitely far away! This dilemma is one expression of **Olbers's paradox**, a problem known since the time of Kepler but brought to the attention of the general public by Heinrich Olbers (1758–1840). If we restrict our attention to the Milky Way Galaxy only, the solution rests in its finite size and nonconstant stellar number density. However, the resolution of Olbers's paradox is not as simple when applied to the universe as a whole, as we will learn in Chapter 29.

The modern process of gathering star count data involves the automated use of CCD detectors to determine $\bar{N}_M$ or A_M . Traditionally, these data are then combined with stellar number densities in the neighborhood of the Sun to estimate the number density of stars of a given spectral type in other regions of the Galaxy. More recently, an iterative computer modeling approach has been used with some success. A general model, based in part on observations of other galaxies believed to be similar to the Milky Way (see Section 25.2), is superimposed on the data; through successive iterations, the density function, the amount of interstellar extinction, and variations in composition with position are fine-tuned until a satisfactory match to the original data is obtained. However, given the complexity of the data and analysis, significant uncertainties remain in our picture of stellar distributions within the Milky Way.

24.2 ■ THE MORPHOLOGY OF THE GALAXY

Armed with numerous examples of other galaxies beyond the Milky Way, together with the data obtained from star counts and information gathered from various distance indicators, abundance analyses, and so on, astronomers have been able to piece together a model of the structure of our own Galaxy. However, it is important to point out that many of the details of the model remain uncertain and may even change in significant ways as more information becomes available. With that caveat, we will attempt to describe our current understanding of the general morphology of the Galaxy in this section, leaving details of its complex motions and fascinating nucleus until later in the chapter.

Distance to the Galactic Center

Just as Herschel and Kapteyn had believed, the Milky Way is a galaxy that possesses a disk of stars, of which the Sun is one member. However, as Shapley had suspected, the Sun does not reside near the center of the disk but is actually located roughly one-third of the way out from the middle. As viewed from Earth, the center of the disk is in the direction of the constellation of Sagittarius corresponding to a very compact emission source known as Sgr A* (see Section 24.4) at the J2000.0 equatorial coordinates

$$\alpha_{\text{Sgr A}^*} = 17^{\text{h}}45^{\text{m}}40.0409^{\text{s}} \quad (24.6)$$

$$\delta_{\text{Sgr A}^*} = -29^{\circ}00'28.118''. \quad (24.7)$$

Face-on and edge-on diagrams of the Galaxy are given in Figs. 24.5 and 24.6, respectively. Some of the details of the Galaxy's components (to be discussed later) are presented in Table 24.1.

The Sun's distance from the center of the Galaxy, known as the **solar Galactocentric distance**, R_0 , has been revised downward many times since Shapley's first estimate of 15 kpc. In 1985 the International Astronomical Union (IAU) recommended a standardized value of $R_0 = 8.5$ kpc for the purpose of allowing direct comparisons of Galactic structure among various researchers. (R_0 is often used to normalize other distances in the Galaxy.) However, a number of studies have found that the value of R_0 is about 8 kpc. For example, in 2003 Frank Eisenhauer and colleagues determined a value of 7.94 ± 0.42 kpc based on astrometric and spectroscopic measurements of S2, the closest star to the Galactic center. Given the remaining uncertainties in the distance to the Galactic center, and given the



FIGURE 24.5 An artist's depiction of the Milky Way Galaxy seen face-on. The shapes of the spiral arms and the length of the bar associated with the central bulge are based on currently available data. The position of the Sun is shown. [Courtesy of NASA/JPL-Caltech/R. Hurt (SSC).]

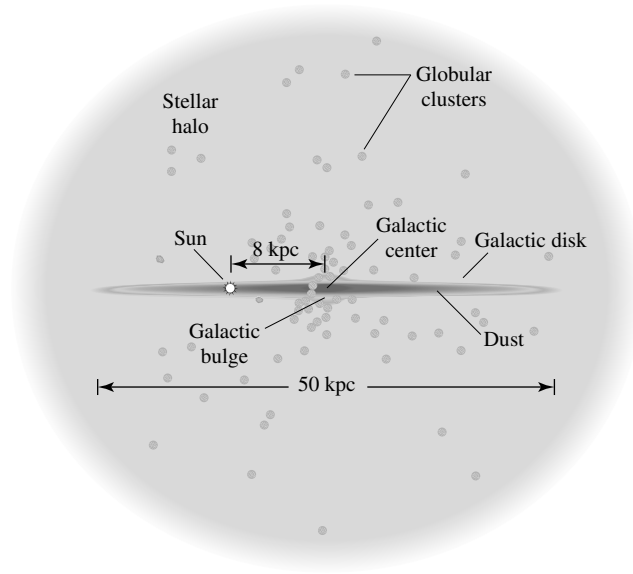


FIGURE 24.6 An edge-on diagram of the Galaxy, not strictly to scale; see Table 24.1.

common use of 8 kpc in the literature, in this text we will adopt a value of

$$R_0 = 8 \text{ kpc} \quad (24.8)$$

for the solar Galactocentric distance.

The full diameter of the disk, including the dust, gas, and stars, is believed to be roughly 50 kpc, with estimates ranging from 40 to 50 kpc. Furthermore, it appears that the disk may not be completely cylindrically symmetric. Rather, the disk may be somewhat elliptical, with a ratio of the lengths of the minor and major axes of about 0.9. The Sun is probably located near the major axis of the disk. The **solar circle** is defined to be a perfect circle of radius R_0 .

The Structure of the Thin and Thick Disks

The disk is actually composed of two major components. The **thin disk**, which is composed of relatively young stars, dust, and gas, has a vertical scale height of $z_{\text{thin}} \simeq 350 \text{ pc}$ and is the region of current star formation (recall that one scale height is the distance over which the number density decreases by e^{-1}). A portion of the thin disk (sometimes referred to as the young thin disk) also corresponds to the central plane of the Galactic dust and gas distribution; it has a scale height of perhaps 90 pc, although some researchers have found a scale height as small as 35 pc. The **thick disk**, which is probably an older population of stars, has a scale height of approximately $z_{\text{thick}} \simeq 1000 \text{ pc}$. The number of stars per unit volume in the thick disk is only about 8.5% of that in the thin disk at the Galactic midplane. When the thin and thick disks are combined, empirical fits to the stellar number density derived from star count data give

$$n(z, R) = n_0 \left(e^{-z/z_{\text{thin}}} + 0.085 e^{-z/z_{\text{thick}}} \right) e^{-R/h_R}, \quad (24.9)$$

TABLE 24.1 Approximate Values for Various Parameters Associated with the Components of the Milky Way Galaxy. Definitions and details are discussed in the text.

Disks			
	Neutral Gas	Thin Disk	Thick Disk
M ($10^{10} M_{\odot}$)	0.5^a	6	0.2 to 0.4
L_B ($10^{10} L_{\odot}$) ^b	—	1.8	0.02
M/L_B ($M_{\odot}/L_{\odot}$)	—	3	—
Radius (kpc)	25	25	25
Form	e^{-z/h_z}	e^{-z/h_z}	e^{-z/h_z}
Scale height (kpc)	< 0.1	0.35	1
σ_w (km s^{-1})	5	16	35
[Fe/H]	$> +0.1$	-0.5 to $+0.3$	-2.2 to -0.5
Age (Gyr)	$\lesssim 10$	8^c	10^d
Spheroids			
	Central Bulge ^e	Stellar Halo	Dark-Matter Halo
M ($10^{10} M_{\odot}$)	1	0.3	$190^{+360}_{-170}{}^f$
L_B ($10^{10} L_{\odot}$) ^b	0.3	0.1	0
M/L_B ($M_{\odot}/L_{\odot}$)	3	~ 1	—
Radius (kpc)	4	> 100	> 230
Form	boxy with bar	$r^{-3.5}$	$(r/a)^{-1} (1 + r/a)^{-2}$
Scale height (kpc)	0.1 to 0.5^g	3	170
σ_w (km s^{-1})	55 to 130^h	95	—
[Fe/H]	-2 to 0.5	< -5.4 to -0.5	—
Age (Gyr)	< 0.2 to 10	11 to 13	~ 13.5

^a $M_{\text{dust}}/M_{\text{gas}} \simeq 0.007$.^b The total luminosity of the Galaxy is $L_{B,\text{tot}} = 2.3 \pm 0.6 \times 10^{10} L_{\odot}$,
 $L_{\text{bol,tot}} = 3.6 \times 10^{10} L_{\odot}$ ($\sim 30\%$ in IR).^c Some open clusters associated with the thin disk may exceed 10 Gyr.^d Major star formation in the thick disk may have occurred 7–8 Gyr ago.^e The mass of the black hole in Sgr A* is $M_{\text{bh}} = 3.7 \pm 0.2 \times 10^6 M_{\odot}$.^f $M = 5.4^{+0.2}_{-3.6} \times 10^{11} M_{\odot}$ within 50 kpc of the center.^g Bulge scale heights depend on age of stars: 100 pc for young stars, 500 pc for old stars.^h Dispersions increase from 55 km s^{-1} at 5 pc to 130 km s^{-1} at 200 pc.

where z is the vertical height above the midplane of the Galaxy, R is the radial distance² from the Galactic center, $h_R > 2.25$ kpc is the disk scale length, and $n_0 \sim 0.02$ stars pc^{-3} for the absolute magnitude range $4.5 \leq M_V \leq 9.5$. It should be pointed out that the relative density coefficient, the scale heights, and the disk scale length are all somewhat uncertain;

²In general, we reserve the use of R to denote the cylindrical coordinate radius within the disk and the use of r to represent the spherical coordinate radius, both measured from the Galactic center.

different researchers give somewhat different values for these parameters. The Sun is a member of the thin disk and is currently located about 30 pc above the midplane.

The **luminosity density** (the luminosity per unit volume) of the thin disk is often modeled with the functional form

$$L(R, z) = L_0 e^{-R/h_R} \operatorname{sech}^2(z/z_0), \quad (24.10)$$

where

$$\operatorname{sech}(z/z_0) = \frac{2}{e^{z/z_0} + e^{-z/z_0}}$$

is the hyperbolic secant function. For the thin disk $z_0 = 2z_{\text{thin}}$ and $L_0 \simeq 0.05 L_\odot \text{ pc}^{-3}$.

The Age–Metallicity Relation

The thin and thick disks not only are identifiable by separate scale heights and stellar number densities but are further distinguished by the chemical compositions and kinematic properties of their members. We will discuss composition effects now but will delay our discussion of kinematics until the next section.

As we noted in Section 13.3, stars are generally classified according to the relative abundance of heavier elements; Population I stars are metal-rich, with $Z \sim 0.02$, Population II stars are metal-poor, with $Z \sim 0.001$, and Population III stars are essentially devoid of metals, with $Z \sim 0$. In reality, a wide range of metallicities exists in stars. At one end are the extreme Population I stars, and on the other, the hypothetical Population III stars (if they still exist). Between Population I and Population II stars are the **intermediate** (or, alternatively and suggestively, **disk**) population stars.

To more carefully quantify the important parameter of composition, the *ratio* of iron to hydrogen has become almost universally adopted by researchers because iron lines are generally readily identifiable in stellar spectra. During a supernova detonation (particularly of Type Ia), iron is ejected, enriching the interstellar medium. New stars can then be created with a greater abundance of iron in their atmospheres than in their predecessors. As a result, iron content should correlate with stellar age, the youngest, most recently formed stars having the highest relative abundance of iron. The iron-to-hydrogen ratio in the atmosphere of a star is compared with the Sun's value through the expression

$$[\text{Fe}/\text{H}] \equiv \log_{10} \left[\frac{(N_{\text{Fe}}/N_{\text{H}})_{\text{star}}}{(N_{\text{Fe}}/N_{\text{H}})_{\odot}} \right], \quad (24.11)$$

a quantity often referred to as the **metallicity**, which was first introduced in Section 23.1. Stars with abundances identical to the Sun's have $[\text{Fe}/\text{H}] = 0.0$, less-metal-rich stars have negative values, and more-metal-rich stars have positive values. Values ranging from -5.4 for old, extremely metal-poor stars to about 0.6 for young, extremely metal-rich stars have been measured in our Galaxy. According to studies of the main-sequence turn-off points in clusters (both globular and galactic), metal-rich stars tend to be younger than metal-poor ones of similar spectral type. The apparent correlation between age and composition is referred to as the **age–metallicity relation**.

However, in many situations the correlation between age and $[\text{Fe}/\text{H}]$ may not be as reliable as first believed. For example, significant numbers of Type Ia supernovae do not appear until roughly 10^9 years after star formation begins, and since Type Ia supernovae are responsible for most of the iron production, iron is not available in large quantities to enrich the interstellar medium. Furthermore, mixing of the interstellar medium after a SN Ia event may not be complete. In other words, a local region of the ISM may become enriched in iron after 10^9 years, while another region may not experience the same level of enrichment. Therefore, according to the age–metallicity relation, the iron-rich region would subsequently produce stars that appear younger, when in fact both regions are the same age.

A second measure of ISM enrichment (and age) is based on $[\text{O}/\text{H}]$, defined in a way completely analogous to Eq. (24.11). Since core-collapse supernovae appear after only 10^7 years following the onset of star formation (recall that they are the result of massive-star evolution) and they produce a higher abundance of oxygen relative to iron, $[\text{O}/\text{H}]$ may also be used to determine the ages of Galactic components; some astronomers use $[\text{O}/\text{Fe}]$ for the same purpose.

To complicate matters further, accurate age estimates depend critically on precise values for distance moduli, which are needed to determine cluster main-sequence turn-off points (Section 13.3); errors of 0.1 in distance moduli lead to 10% errors in age. Needless to say, the important task of Galactic age determinations remains a major challenge in astronomy.

Age Estimates of the Thin and Thick Disks

In the thin disk, typical values for the iron–hydrogen metallicity ratio are in the range $-0.5 < [\text{Fe}/\text{H}] < 0.3$, while for the majority of stars in the thick disk, $-0.6 < [\text{Fe}/\text{H}] < -0.4$ is more characteristic, although some thick-disk members may have metallicities at least as low as $[\text{Fe}/\text{H}] \sim -1.6$.

According to various age determinations, the stellar members of the thin disk are probably significantly younger than their thick-disk counterparts. It appears that star formation began in the thin disk about 8 Gyr ago and is ongoing today. This conclusion is supported by the observations of white dwarf stars in the thin disk and theoretical estimates of their cooling times (Section 16.5). There is also some evidence that star formation in the thin disk may not have been continuous over time but may have come in bursts with intervening gaps of several billion years. On the other hand, star formation in the thick disk appears to have predated the onset of star formation in the thin disk by two to three billion years. It is generally believed that the episode of thick-disk star formation spanned the time interval between 10 and 11 Gyr ago.

Mass-to-Light Ratios

Based on data from star counts and orbital motions, the estimated stellar mass of the thin disk is roughly $6 \times 10^{10} M_{\odot}$ with another $0.5 \times 10^{10} M_{\odot}$ of dust and gas. Furthermore, the luminosity in the blue-wavelength band is $L_B = 1.8 \times 10^{10} L_{\odot}$. When the first of these parameters is divided by the second, the resulting **mass-to-light ratio** is $M/L_B \approx 3 M_{\odot}/L_{\odot}$. This quantity gives us information about the kinds of stars responsible for the generation of the light.

We learned from Fig. 7.7 and Problem 10.27 that along the main sequence, a star's luminosity depends rather strongly on its mass, with

$$\frac{L}{L_{\odot}} = \left(\frac{M}{M_{\odot}} \right)^{\alpha}, \quad (24.12)$$

where $\alpha \simeq 4$ above about $0.5 M_{\odot}$ and $\alpha \simeq 2.3$ for less massive stars. Assuming that most of the stars in the disk are main-sequence stars, an “average” stellar mass can be estimated. Substituting the observed mass-to-light ratio and solving for the mass, we have

$$\langle M \rangle = 3^{1/(1-\alpha)} M_{\odot}.$$

Assuming that $\alpha \simeq 4$, we find that $\langle M \rangle \simeq 0.7 M_{\odot}$. Apparently the total luminosity of the disk is dominated by stars somewhat less massive than the Sun. This should not be surprising since, as we saw in Fig. 12.12, the initial mass function indicates that many more low-mass than high-mass stars are created out of the interstellar medium, which is consistent with the fact that dwarf M stars are the most common class of stars in the vicinity of the Sun.

The *B*-band luminosity of the thick disk is about $2 \times 10^8 L_{\odot}$, or 1% of the value of the thin disk (which explains why the thick disk has been so difficult to detect). The mass of the thick disk is probably about 2 to $4 \times 10^9 M_{\odot}$, or approximately 3% of the thin-disk mass.

Spiral Structure

Significant structure exists within the disk itself. When neutral hydrogen clouds or relatively young objects such as O and B stars, H II regions, and galactic (open) clusters are used as tracers of Galactic structure, a **spiral structure** emerges, giving the disk the appearance of a pinwheel. When other galaxies that possess distinct disks are observed in blue light (the dominant visible-wavelength band of hot, bright, young, high-mass main-sequence stars), these galaxies often exhibit similarly beautiful spiral structure. One such example is the great spiral galaxy of Andromeda, shown in Fig. 24.7. However, when the galaxies are viewed in the red light characteristic of older, low-mass stars, the spiral structure is less pronounced. It appears that spirals are associated with ongoing star formation and that older stars have had ample time to drift out of the spiral pattern (see Section 25.3). As we can see in Fig. 24.8, the Sun seems to be located close to, but not actually in, one of the spiral arm features known as the **Orion–Cygnus arm**, or simply the **Orion arm** (this feature is also known as the **Orion spur** since it is probably not a full spiral arm structure). Spiral arms get their names from the constellations in which they are observed.

The interstellar gas and dust clouds that plagued Kapteyn's attempts at determining the overall structure of the Galaxy and that are clearly evident in Fig. 24.1 are primarily located near the midplane and found preferentially in the spiral arms. If it were possible to view our Galaxy from a vantage point outside the disk but along the plane, it would probably appear similar to NGC 891,³ the galaxy shown in Fig. 24.9.

³NGC is a designation indicating that the galaxy is a member of the *New General Catalog*, discussed on page 940.

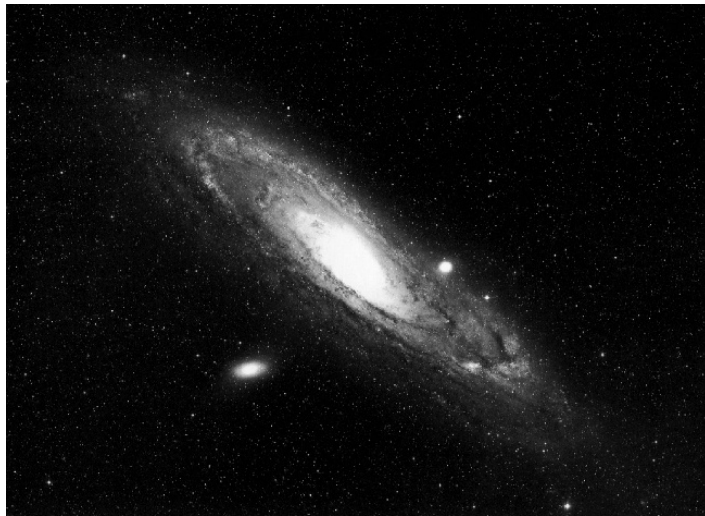


FIGURE 24.7 The Andromeda galaxy (also known as M31 or NGC 224) is a spiral galaxy believed to be much like our own. M31 is located 770 kpc from the Milky Way Galaxy. (From Sandage and Bedke, *The Carnegie Atlas of Galaxies*, The Carnegie Institution of Washington, Washington, D.C., 1994.)

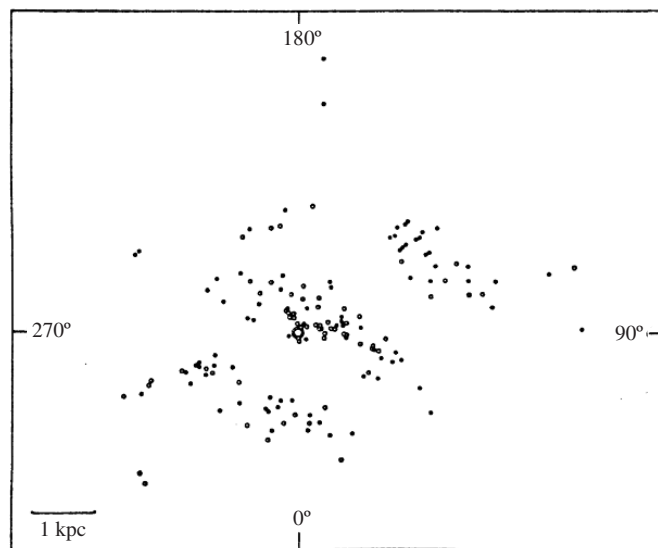


FIGURE 24.8 The spatial distribution of young Galactic clusters and H II regions reveals the presence of spiral arms within the disk of the Galaxy. The Sun is shown at the origin of the diagram (the intersection of the degree tick marks). The Sun is located near the Orion–Cygnus arm. [Figure adapted from Becker and Fenkart, *The Spiral Structure of Our Galaxy*, Becker and Contopoulos (eds.), D. Reidel Publishing Company, Dordrecht, 1970.]

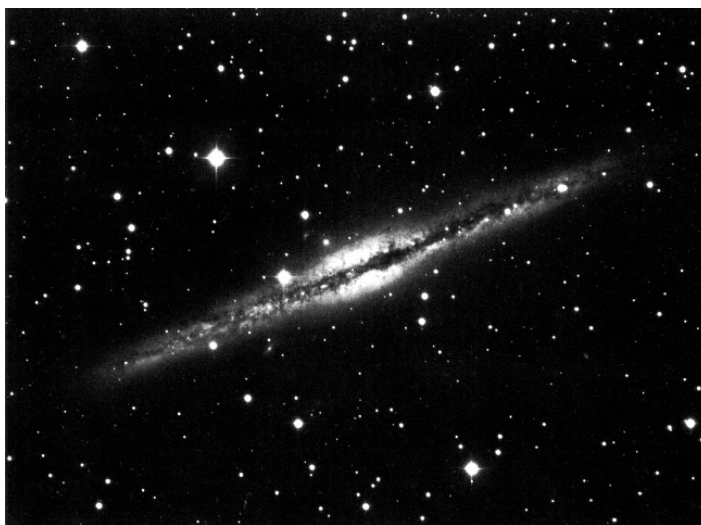


FIGURE 24.9 NGC 891, seen edge-on, clearly shows a thin dust band in the plane of the disk. The Milky Way Galaxy probably appears much like NGC 891 when viewed from a distant vantage point. (From Sandage and Bedke, *The Carnegie Atlas of Galaxies*, The Carnegie Institution of Washington, Washington, D.C., 1994.)

Interstellar Gas and Dust

As we discussed in Section 12.1, gas and dust clouds exist in the Milky Way with a range of masses, temperatures, and densities. It is from these clouds that new stars are ultimately formed. Astronomers have been able to map out the overall distribution of dust and gas within the Milky Way by measuring the effects of obscuration by and emissivity of dust, as well as the location of 21-cm H I emission, and by using the CO molecule as a tracer of H₂. Molecular hydrogen and cool dust are found predominantly in the regions 3 to 8 kpc and 3 to 7 kpc from the Galactic center, respectively (i.e., inside the solar circle), while atomic hydrogen can be found in the region from 3 kpc out to the edge of the Galactic disk (25 kpc). It appears that H₂ and the dust are most tightly confined to the plane of the Galaxy, with vertical scale heights above or below the midplane of perhaps 90 pc or less. This is only about 25% of the value for stars in the thin disk and on the order of 9% of the scale height of thick-disk stars. In the region near the Sun, the scale height for atomic hydrogen is approximately 160 pc. The total mass of H I is estimated to be $4 \times 10^9 M_{\odot}$ and the mass of H₂ is approximately $10^9 M_{\odot}$. In the solar neighborhood the total mass density of gas is $0.04 M_{\odot} \text{ pc}^{-3}$, of which atomic hydrogen accounts for approximately 77%, molecules contribute 17%, and ions add an additional 6%.

At distances beyond 12 kpc from the center of the Galaxy, the scale height of H I increases dramatically, reaching a value of more than 900 pc. In addition, the distribution of H I in the outer reaches of the Galaxy is no longer strictly confined to the plane but, rather, exhibits a well-defined **warp** that reaches a maximum angle of deviation from the plane of 15°. A map of the H I distribution at a Galactic radius of 13.6 kpc is shown in Fig. 24.10. The

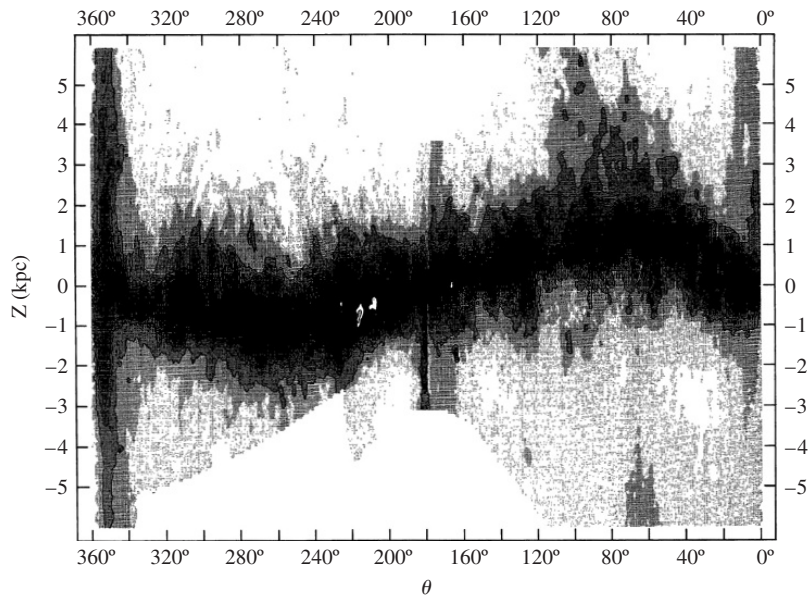


FIGURE 24.10 The H I warp in the Milky Way at a distance of 13.6 kpc from the Galactic center. The Galaxy's midplane is located at $z = 0$, and the direction toward the center of the Milky Way is at 0° . (Figure adapted from Burton and de Lint Hekker, *Astron. Astrophys. Suppl. Ser.*, 65, 427, 1986.)

line of sight toward the center of the Milky Way is labeled as 0° in the figure. Warped H I distributions such as the one observed in our own Galaxy appear to be common features in other spiral galaxies as well, including Andromeda (in some spiral galaxies the warp angle can reach 90°). Although we do not yet fully understand the dynamics that lead to the creation of warps (they do not seem to be the result of simple gravitational perturbations from one or more external galaxies, for example), they do seem to be associated with the distribution of mass in the outer regions of the Galaxy, beyond where most of the luminosity is produced.

Hydrogen clouds can also be found at high latitudes. Although some of these clouds have positive radial velocities, implying that they are moving away from the disk, the majority possess large negative radial velocities (up to 400 km s^{-1} or more), as measured by their 21-cm emission. There appear to be two types of sources responsible for these **high-velocity clouds**. In one process, clouds of gas ejected from supernovae are driven to large values of z , where they eventually cool and rain back down onto the Galactic plane. This suggestion is known as the **Galactic fountain model**. It also appears that the Galaxy is accreting gas from intergalactic space, as well as from a number of its small satellite galaxies; hence the predominance of negative-radial-velocity clouds.

In addition, a very hot, tenuous gas exists at distances up to or exceeding 70 kpc from the Galactic center. The **Far Ultraviolet Spectroscopic Explorer** (FUSE) has been able to detect O VI absorption lines in the spectra of distant extragalactic sources and halo stars produced when the light passes through the gas in the Galactic halo. Using O VI as a tracer of the hydrogen gas, the strengths of the O VI absorption lines imply a number density of

hydrogen of $n_H \sim 10^{-11} \text{ m}^{-3}$. Assuming that the distribution is approximately spherical with a radius of $R \sim 70 \text{ kpc}$ leads to an estimate of the mass of the gas of $M \simeq 4 \times 10^8 M_\odot$. If the gas distribution is somewhat flattened, the total mass could exceed $10^9 M_\odot$. In order to support the gas against gravitational collapse, the gas temperature must be very hot. It is estimated that the temperature of this hot, tenuous gas is in excess of 10^6 K , with the high gas temperature perhaps being due to collisions between infalling gas and existing gas in the Galaxy. The gas is sometimes referred to as a **coronal gas**, suggestive of its high temperature.

The Disruption of Satellite Galaxies

Another unusual high-latitude feature is the **Magellanic Stream**, a narrow band of H I emission stretching more than 180° across the sky and trailing the Southern Hemisphere's Magellanic Clouds (recall that the Large and Small Magellanic Clouds are small satellite galaxies of the Milky Way; the LMC and SMC are located 52 and 61 kpc from Earth, respectively).⁴ The Magellanic Stream appears to be the result of a tidal encounter of the Magellanic Clouds with the Milky Way some 200 Myr ago. It has also been suggested that the Magellanic Stream may owe some of its structure to the interaction with our Galaxy's hot coronal gas.

Other satellite galaxies have also tidally interacted with the Milky Way, and still others are currently doing so. For example, in 1995 Rodrigo Ibata and his colleagues announced the discovery of a previously unknown dwarf spheroidal galaxy in Sagittarius. At a distance of only 24 kpc from Earth and 16 kpc from the center of the Milky Way, the Sagittarius dwarf spheroidal is the closest galaxy to Earth. It is clearly elongated with the long axis directed toward the center of the Galaxy and, with a radial velocity of 140 km s^{-1} , has had only a few orbital encounters with the Milky Way. Evidently the Sagittarius dwarf spheroidal galaxy, along with its globular clusters, is being incorporated into the Milky Way Galaxy.

Using the **2-Micron All Sky Survey** (2MASS) catalog, researchers have also identified an overdensity of stars in the constellation Canis Major near the plane of the Milky Way. A group of globular clusters and open clusters are associated with this overdensity in both position and radial velocity. This feature strongly suggests that another dwarf satellite galaxy was integrated into the Milky Way in the past and may now be a part of the thick disk.

The unusual globular cluster ω Centauri also seems to be the remnant of a dwarf galaxy that has been subsumed by the Milky Way. ω Cen is the largest and brightest globular cluster visible from Earth and has an unusually high surface brightness. It appears that this globular cluster is the stripped core of another former satellite galaxy. (It has been suggested that the globular clusters M54 and NGC 2419 were also once dwarf galaxies that suffered the same fate as ω Cen.)

The Galactic Bulge

Although the vertical scale height of the thin disk is near 350 pc in the vicinity of the Sun, that value increases somewhat toward the inner regions of the Galaxy, where the disk meets the **Galactic bulge**. The bulge is not simply an extension of the disk but is an independent

⁴The LMC contains 30 Dorodus and SN 1987A, both discussed in Chapter 15.

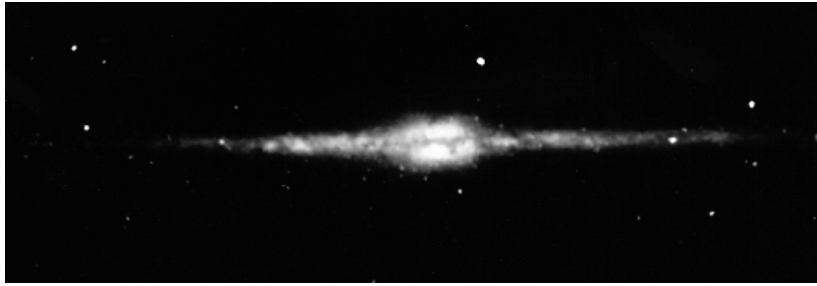


FIGURE 24.11 An infrared view of the Galaxy, as seen by COBE. The image was produced from observations at $1.2\ \mu\text{m}$, $2.2\ \mu\text{m}$, and $3.4\ \mu\text{m}$, and extends 96° either side of the Galactic center. (Courtesy of the COBE Science Working Group and NASA's Goddard Space Flight Center.)

component of the Galaxy. The mass of the bulge is believed to be roughly $10^{10}\ M_\odot$ and its B -band luminosity is near $3 \times 10^9\ L_\odot$. This gives a mass-to-light ratio of $3\ M_\odot/L_\odot$, a value comparable to that found for the thin disk.

The boxy (or elongated) bulge is evident in the **COsmic Background Explorer satellite** (COBE) image shown in Fig. 24.11. The image, which was produced by combining observations at $1.2\ \mu\text{m}$, $2.2\ \mu\text{m}$, and $3.4\ \mu\text{m}$, should be compared to the optical picture of NGC 891, shown in Fig. 24.9. Using the COBE data, together with observations of RR Lyraes and K and M giants, we find that the variation in the number density of stars in the bulge corresponds to a vertical scale height that ranges from 100 pc to 500 pc, depending on the ages of the stars used to make the determination; younger stars yield smaller scale heights.

The surface brightness, I , of the bulge (measured in units of $L_\odot\ \text{pc}^{-2}$) exhibits an approximate radial dependence of the form

$$\log_{10} \left[\frac{I(r)}{I_e} \right] = -3.3307 \left[\left(\frac{r}{r_e} \right)^{1/4} - 1 \right], \quad (24.13)$$

which is often referred to as a **$r^{1/4}$ law** [the law was first formulated by Gerard de Vaucouleurs (1918–1995) in 1948 and is also referred to as a **de Vaucouleurs profile**]. r_e is a reference radius (called the **effective radius**) and I_e is the surface brightness at r_e . Formally, r_e is defined to be that radius within which one-half of the bulge's light is emitted. At the infrared wavelength of $12\ \mu\text{m}$, star count data from the **InfraRed Astronomical Satellite** (IRAS) suggests an effective radius of about 0.7 kpc. Similar results were found by COBE. (We will find that the de Vaucouleurs profile is a general distribution for relaxed, spheroidal systems.)

A serious difficulty in observing the properties of the bulge rests in the large amount of extinction at visible wavelengths due to the dust between the Sun and the Galactic center. The total amount of extinction within several degrees of the center can be more than 30 magnitudes. However, a number of lines of sight exist for which the amount of extinction is minimal. The most well known of these is **Baade's window**, which Walter Baade (1893–

1960) discovered in 1944 while observing the globular cluster NGC 6522. Baade realized that by observing in that region of the sky, he was able to see RR Lyraes that were actually beyond the Galactic center. Baade's window is 3.9° below the Galactic center, and the line of sight passes within 550 pc of the center. It is believed that NGC 6522 is located within the bulge.

From the observational evidence, the chemical abundances of stars in the bulge vary significantly, ranging from quite metal-poor to very metal-rich; $-2 < [\text{Fe}/\text{H}] < 0.5$. In fact, based on the chemical abundances, it appears that three somewhat distinct age groupings exist in the central bulge. One set of stars appear to be very young, with ages less than 200 Myr, a second set has ages between 200 Myr and 7 Gyr, and the third set tends to be older than 7 Gyr (perhaps up to 10 Gyr or older).

In a trend that on initial reflection appears to be counterintuitive, the oldest stars in the bulge tend to have the highest metallicities, while the youngest stars have a fairly uniform distribution of metallicities across the range from -2 to 0.5 . This pattern is probably due to a burst of massive star formation when the Galaxy was young. Apparently core-collapse supernovae enriched the interstellar medium early in the life of the bulge, implying that subsequent generations of stars contained an enhanced abundance of heavier elements. The more uniform distribution of metallicity in recent generations of stars could be the result of fresh, infalling material. It is this type of complication that must be considered carefully when employing the age–metallicity relation discussed on page 885.

The Milky Way's Central Bar

Although it was originally thought to be essentially spheroidal in nature, a number of observing campaigns and database studies have determined that the bulge contains a distinct **bar**.⁵ The Milky Way's central bar is clearly depicted in the artist's drawing of the Galaxy shown in Fig. 24.5. The bar has a radius (one-half length) from the galactic center of 4.4 ± 0.5 kpc and is oriented at an angle of $\phi = 44^\circ \pm 10^\circ$ with respect to the line-of-sight angle from Earth to the Galactic center. It also seems that the bar is somewhat thicker in the plane of the Galaxy than in the z direction; the dimension ratios are approximately 1:0.5:0.4. (As we will learn in Chapter 25, bars are dynamically stable structures and common features of many spiral galaxies.)

The 3-kpc Expanding Arm

A unique feature in the inner regions of the Galaxy that is most easily observed at the 21-cm wavelength of H I is the **3-kpc expanding arm**, a gas cloud that is moving toward us at roughly 50 km s^{-1} . Once believed to be the product of a gigantic explosion in the center of the Galaxy, the rapidly moving structure is now thought to be a consequence of the presence of the stellar bar. Rather than being driven away from the center in an explosive event that would require an unrealistic 10^{52} J of energy, the gas cloud is merely in a very elliptical orbit about the Galactic center resulting from gravitational perturbations from the bar.

⁵One such database study is the analysis of the GLIMPSE (Galactic Legacy Mid-Plane Survey Extraordinaire) point source catalog of some 30 million infrared sources in the direction of the inner Galaxy, produced by the Spitzer Space Telescope. The Galactic bulge is much more transparent to infrared wavelengths than to visible wavelengths.

The Stellar Halo and Globular Cluster System

The last luminous component of the Galaxy is the **stellar halo** (or simply the halo), composed of the globular clusters and those **field stars** (stars that are not members of clusters) that have large velocity components perpendicular to the Galactic plane. These field stars are often referred to as **high-velocity stars** since, as we will learn in the next section, their velocity components differ significantly from those of the Sun. Most of the globular clusters and the high-velocity stars can reach positions that are far above or below the plane of the Galaxy.

Although it appeared to Shapley that all of the known globular clusters were distributed nearly spherically about the Galactic center, it has become apparent that two distinct spatial distributions exist, delineated by metallicity. Older, metal-poor clusters whose members have $[\text{Fe}/\text{H}] < -0.8$ belong to an extended spherical halo of stars, while younger clusters with $[\text{Fe}/\text{H}] > -0.8$ form a much flatter distribution and may even be associated with the thick disk. The spatial distribution of the two metallicity groups is shown in Fig. 24.12. The notable exception is the well-studied globular cluster 47 Tucana (47 Tuc; also known as NGC 104), which is located 3.2 kpc below the Galactic plane and has an unusually high metallicity of $[\text{Fe}/\text{H}] = -0.67$. Some astronomers have argued that 47 Tuc is a member of the halo population, while others consider 47 Tuc to be a member of the thick disk. A color–magnitude diagram of 47 Tuc was presented in Fig. 13.20(a).

Our Galaxy is known to contain at least 150 globular clusters with distances from the center of the Milky Way ranging from 500 pc to 120 kpc. The youngest globular clusters appear to be about 11 Gyr old, and the oldest are probably a little over 13 Gyr old.⁶ It now appears that a significant age spread of two billion years or so exists between the youngest and oldest members of the halo.

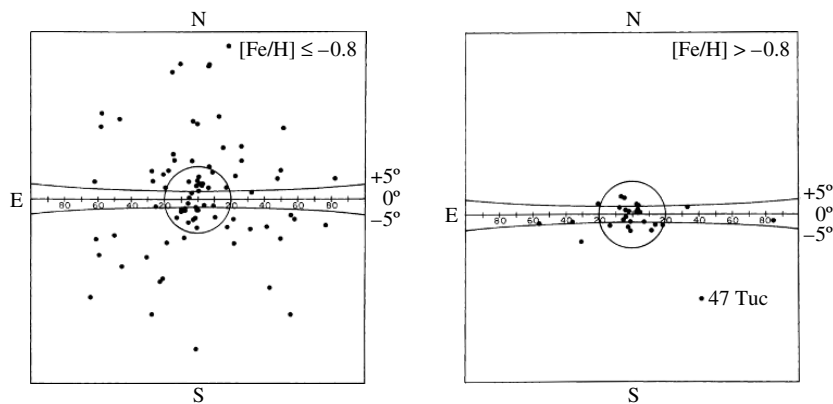


FIGURE 24.12 Metal-poor globular clusters form a nearly spherical distribution about the Galactic center, while more metal-rich clusters are found preferentially near the plane of the Galaxy, possibly associated with the thick disk. (Figure adapted from Zinn, *Ap. J.*, 293, 424, 1985.)

⁶Krauss and Chaboyer, *Science*, 299, 65, 2003, find a best-fit age for the globular clusters of 12.6 Gyr, with lower and upper limits of 10.4 Gyr and 16 Gyr, respectively, at 95% confidence levels. Gratton et al., *Aston. Astrophys.*, 408, 529, 2003, find ages for NGC 6397, NGC 6752, and 47 Tuc of 13.9 ± 1.1 Gyr, 13.9 ± 1.1 Gyr, and 11.2 ± 1.1 Gyr, respectively. For comparison, the age of the universe is 13.7 ± 0.2 Gyr.

Although 144 of the globular clusters are found within 42 kpc of the Galactic center, 6 globular clusters have been found between 69 and 123 kpc of the Galactic center. Some astronomers have suggested that those six most distant clusters may have been captured by the Milky Way or may be dwarf spheroidal galaxies, much as ω Cen and the Sagittarius dwarf galaxy may have been. If we do not include these very remote objects, the metal-poor clusters seem to be confined to a halo with a radius of approximately 42 kpc. However, the detection of extremely distant and luminous field stars suggests that a stellar halo radius of 50 kpc may be more appropriate.

The number density profile of the metal-poor globular clusters and the field stars in the halo has the form

$$n_{\text{halo}}(r) = n_{0,\text{halo}}(r/a)^{-3.5}, \quad (24.14)$$

where $n_{0,\text{halo}} \simeq 4 \times 10^{-5} \text{ pc}^{-3}$ is roughly 0.2% of the thin disk's midplane value (see Eq. 24.9). The scale length (a) of the number density distribution is several thousand parsecs. At visible wavelengths, the effective radius of the halo $r^{1/4}$ law (Eq. 24.13) is $r_e = 2.7 \text{ kpc}$.

Metal-rich globular clusters seem to have many characteristics in common with thick-disk field stars, including spatial distribution. Just like the thick-disk field stars, the vertical scale height of these clusters is on the order of 1 kpc, unlike the much larger scale length of the metal-poor clusters.

However, based on studies that use RR Lyraes as tracers of other field stars in the halo, there is some question about whether field stars actually share the same spatial distribution pattern as the corresponding metal-poor clusters. Instead, the field stars may occupy a volume that is appreciably flattened, perhaps as much as $c/a \sim 0.6$, where c is the minor axis of a spheroid in the direction perpendicular to the Galactic plane, and a is the major axis (see Fig. 24.13). Such a discrepancy in distributions, if real, could have a significant influence on an ultimate model of Galaxy formation and evolution. On the other hand, other

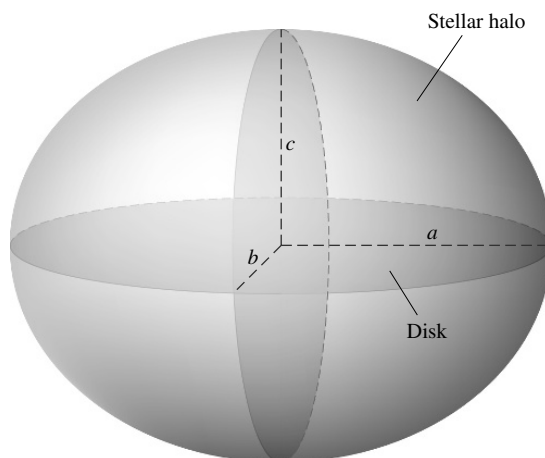


FIGURE 24.13 The general shape of a triaxial spheroidal system, where the three axes of the spheroid are assumed to have $a \geq b \geq c$. The Galactic disk (shown foreshortened by perspective) is roughly circular.

astronomers believe that a single value of $0.8 \leq c/a \leq 0.9$ probably applies to the field stars and clusters alike.

Based largely on star counts, the total *stellar* mass density in the solar neighborhood is approximately $0.05 \text{ M}_\odot \text{ pc}^{-3}$. Of that total, the contribution due to high-velocity halo stars is on the order of 0.2%, or $1 \times 10^{-4} \text{ M}_\odot \text{ pc}^{-3}$. Combining this with the spatial density power law yields a total estimated mass of the stellar halo on the order of $1 \times 10^9 \text{ M}_\odot$, of which roughly 1% is the combined mass of the globular clusters; the remainder is locked up in field stars. The *B*-band luminosity of the halo is estimated to be $1 \times 10^9 \text{ L}_\odot$, giving a mass-to-light ratio on the order of unity (the exact value is quite uncertain).

When all of the components of the Galaxy are considered together, its total *B*-band luminosity is $L_{B,\text{tot}} = 2.3 \pm 0.6 \times 10^{10} \text{ L}_\odot$. However, approximately one-third of the Galaxy's emission is in the infrared part of the electromagnetic spectrum and is primarily associated with interstellar dust. Including IR emissions, the bolometric luminosity of the Galaxy is $L_{\text{bol}} = 3.6 \times 10^{10} \text{ L}_\odot$.

The Dark Matter Halo

When the masses of all of the luminous components of the Galaxy are combined (the thin and thick disks, the interstellar dust and gas, the Galactic bulge, the stellar halo, and the bar), the total mass of luminous matter in the Galaxy is estimated to be $9 \times 10^{10} \text{ M}_\odot$. As we shall see in the next section, this value is in good agreement with the orbital motion of the Sun about the Galactic center, but it does not explain the orbits of stars and gas at Galactocentric distances much greater than R_0 . Apparently there is still another crucial component to the overall structure of our Galaxy. Along with influencing orbital motions, this unseen element of Galactic structure may also be responsible for the generation of the warps seen in the H I distribution near the outer edges of the luminous disk.

This **dark matter halo** seems to be roughly spherically distributed, enveloping the stellar halo and extending out to at least 230 kpc. Based on its gravitational influence over luminous matter, the dark matter halo has an apparent mass distribution of the form

$$\rho(r) = \frac{\rho_0}{(r/a)(1 + r/a)^2}, \quad (24.15)$$

where ρ_0 and a are chosen by fitting the mass distribution in the dark matter halo (estimates of ρ_0 and a will be made in Problem 24.24). This functional dependence behaves as $1/r$ when $r \ll a$ and as $1/r^3$ when $r \gg a$. The mass of the dark matter halo may be as great as $5.4 \times 10^{11} \text{ M}_\odot$ within 50 kpc of the Galactic center and $1.9 \times 10^{12} \text{ M}_\odot$ within 230 kpc of the Galactic center. It appears that the dark matter halo accounts for about 95% of the entire mass of the Galaxy. [The empirical justification for a dark matter density distribution such as Eq. (24.15) will be discussed beginning on page 914.]

The composition of the dark matter halo is still a mystery. It cannot be in the form of interstellar dust, because dust betrays its presence through the extinction of starlight. Furthermore, the dark matter halo cannot be composed of gas, because absorption lines would be apparent when observing halo stars. One possible class of candidates are referred to as **weakly interacting massive particles** (WIMPs). WIMPs would not contribute to the overall luminosity of the Galaxy, but they would affect it through their gravitational

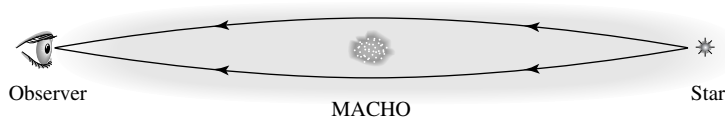


FIGURE 24.14 Gravitational lensing (focusing) of starlight produced by an intervening MACHO.

interactions. In support of a WIMP-dominated Galaxy, theoretical considerations related to the formation and subsequent evolution of the universe suggest that nonbaryonic matter (constituents of matter other than protons, neutrons, and related massive particles) may constitute the majority of the dark matter halo's mass, an issue that will be discussed in more detail in Chapters 29 and 30. It had been suggested that neutrinos (which are leptons) fill the bill, although it now appears that they aren't massive enough to explain the amount of mass that must be present in the dark matter halo. Perhaps currently hypothetical (and undetected) particles are responsible, such as the suggested supersymmetric particles known as **neutralinos** that are implied in some grand unified theories of particle physics.

A competing hypothesis for the possible makeup of dark matter suggests that **massive compact halo objects** (MACHOs) may be responsible. MACHOs that could supply the unseen mass may be in the form of white dwarfs, neutron stars, black holes, or less exotic red dwarfs or brown dwarfs. Some searches for MACHOs have been based on the general-relativistic prediction that starlight is deflected as it passes near a massive object (Section 17.1). If a MACHO is located between a distant star and Earth, the light from the star can be focused with the MACHO acting as a **gravitational lens**, as shown in Fig. 24.14. In 1993, while observing stars in the Large Magellanic Cloud, astronomers detected the telltale signature of temporary brightening that is predicted to occur when a MACHO passes through the line of sight. Figure 24.15 shows the light curves of one event that lasted for 33 days. From a statistical analysis of the small number of such events, it appears that there are too few MACHOs to account for a significant fraction of the mass contained in the dark matter halo.

That conclusion is also supported by searches for white dwarfs and small red dwarf stars, carried out using the Hubble Space Telescope. In those studies, the number of stars detected is too small to constitute a major component of the dark matter in the Galaxy. Based on the faintest (deepest) searches yet conducted at the time of writing, white dwarfs can contribute no more than 10% of the dark matter halo's mass, and red dwarfs no more than 6% of it.

The Galactic Magnetic Field

Like many other astrophysical environments, the Galaxy possesses a magnetic field. The orientation and strength of the field can be measured in a variety of ways, including the Zeeman effect and the polarization of visible- and radio-wavelength electromagnetic radiation by reflection from interstellar grains aligned with the field. It appears that within the disk, the field tends to follow the Galaxy's spiral arms and has a typical strength of 0.4 nT. The field strength in the stellar halo may be an order of magnitude weaker than in the disk, and the field strength near the Galactic center may reach 1 μT (see Section 24.4).

Although the global Galactic magnetic field is quite weak relative to the magnetic field near the surface of Earth (about 50 μT), it probably plays a significant role in the structure

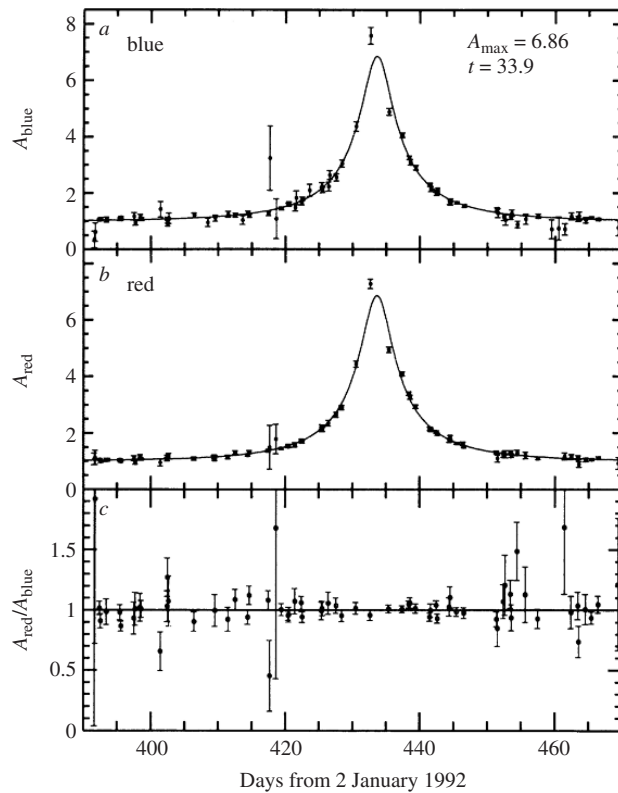


FIGURE 24.15 The light curve of a star in the LMC brightened over a period of 33 days, apparently because a MACHO passed through the line of sight. The data are shown for (a) blue light, (b) red light, and (c) the ratio of blue light to red light. (Figure adapted from Alcock et al., *Nature*, 365, 621, 1993. Reprinted with permission, © 1993, Macmillan Magazines Limited.)

and evolution of the Milky Way. This can be seen by considering its energy density, which appears to be comparable (perhaps equal) to the thermal energy density of gas within the disk (see Problem 24.9).

24.3 ■ THE KINEMATICS OF THE MILKY WAY

One of the great challenges in understanding the nature of our Galaxy lies in determining its internal kinematics. In this section, we will discuss how astronomers have been able to uncover the complexities of motions in the Milky Way from our vantage point in orbit around a moving Sun.

The Galactic Coordinate System

The Galactic midplane is not aligned with the plane of the celestial equator but is inclined at an angle of 62.87° to it. As a result, rather than using the Earth-based equatorial coordi-

nate system, it is more convenient to introduce a new coordinate system when discussing the structure and kinematics of the Milky Way Galaxy. The **Galactic coordinate system** exploits the natural symmetry introduced by the existence of the Galactic disk.

The intersection of the midplane of the Galaxy with the celestial sphere forms what is very nearly a great circle, known as the **Galactic equator**.⁷ This orientation is depicted in Fig. 24.16. **Galactic latitude** (b) and **Galactic longitude** (ℓ) are defined from a vantage point taken to be the Sun, as shown in Fig. 24.17. Galactic latitude is measured in degrees

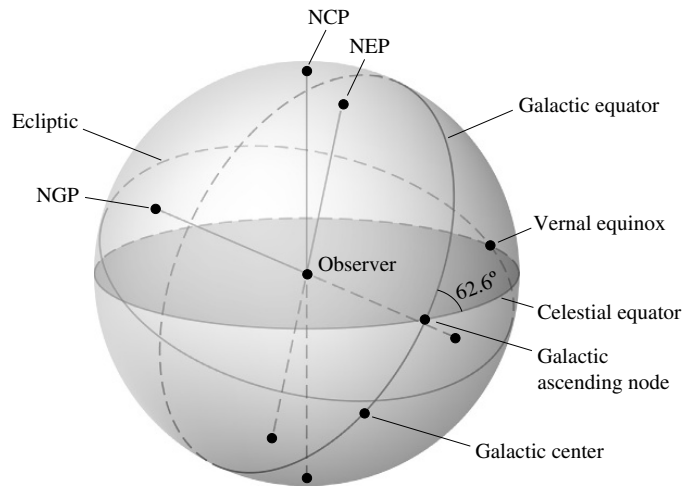


FIGURE 24.16 The relative orientations of the celestial equator, the ecliptic, and the Galactic equator as they appear on the celestial sphere. Note that from the vantage point depicted in the figure, the north celestial pole (NCP), the north ecliptic pole (NEP), and north Galactic pole (NGP) are all on the front of the celestial sphere.

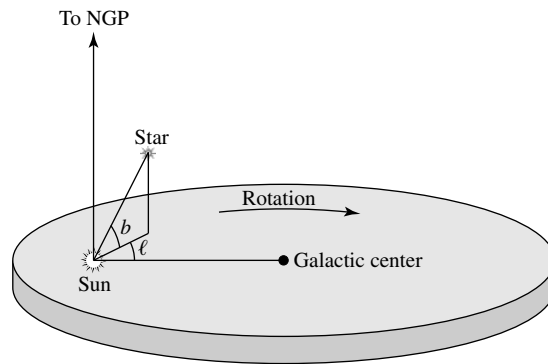


FIGURE 24.17 The definition of the Galactic coordinates ℓ and b . The direction of rotation of the Galaxy is also labeled.

⁷Technically, the Galactic midplane does not exactly trace out a great circle, because the Sun is not precisely in the midplane; however, the deviation is very small. Recall from Section 24.2 that the Sun is located just 30 pc above the plane.

north or south of the Galactic equator along a great circle that passes through the **north Galactic pole**. Galactic longitude (also in degrees) is measured east along the Galactic equator, beginning *near* the Galactic center, to the point of intersection with the great circle used to measure Galactic latitude.

By international convention, the J2000.0 equatorial coordinates of the north Galactic pole ($b = 90^\circ$) are

$$\alpha_{\text{NGP}} = 12^{\text{h}}51^{\text{m}}26.28^{\text{s}}$$

$$\delta_{\text{NGP}} = 27^\circ 7' 41.7'',$$

and the origin of the Galactic coordinate system ($\ell_0 = 0^\circ$, $b_0 = 0^\circ$) corresponds to

$$\alpha_0 = 17^{\text{h}}45^{\text{m}}37.20^{\text{s}}$$

$$\delta_0 = -28^\circ 56' 9.6''.$$

Note that the center of the Galaxy ($\alpha_{\text{Sgr A}^*}$, $\delta_{\text{Sgr A}^*}$) given by Eqs. (24.6) and (24.7) is very close to, but not exactly aligned with, ($\ell_0 = 0^\circ$, $b_0 = 0^\circ$).

Two other useful positions on the sky are also worth specifying in both coordinate systems. The location of the north celestial pole ($\delta_{\text{NCP}} = 90^\circ$), given in J2000.0 Galactic coordinates, is

$$\ell_{\text{NCP}} = 122^\circ 55' 55.2''$$

$$b_{\text{NCP}} = 27^\circ 7' 41.7''.$$

And the intersection of the celestial equator with the Galactic equator moving eastward from negative to positive declination (the *ascending node*) is given in equatorial coordinates by

$$\alpha_{\text{ascending}} = 18^{\text{h}}51^{\text{m}}24^{\text{s}}$$

$$\delta_{\text{ascending}} = 0^\circ$$

and in Galactic coordinates by

$$\ell_{\text{ascending}} = 33.0^\circ$$

$$b_{\text{ascending}} = 0^\circ.$$

The transformations between equatorial and Galactic coordinates involves the application of the methods of spherical trigonometry. To make the transformation from equatorial to Galactic coordinates (assuming epoch J2000.0):

$$\sin b = \sin \delta_{\text{NGP}} \sin \delta + \cos \delta_{\text{NGP}} \cos \delta \cos(\alpha - \alpha_{\text{NGP}}) \quad (24.16)$$

$$\cos b \sin(\ell_{\text{NCP}} - \ell) = \cos \delta \sin(\alpha - \alpha_{\text{NGP}}) \quad (24.17)$$

$$\cos b \cos(\ell_{\text{NCP}} - \ell) = \cos \delta_{\text{NGP}} \sin \delta - \sin \delta_{\text{NGP}} \cos \delta \cos(\alpha - \alpha_{\text{NGP}}). \quad (24.18)$$

For transformations from Galactic to equatorial coordinates (again assuming epoch J2000.0):

$$\sin \delta = \sin \delta_{\text{NGP}} \sin b + \cos \delta_{\text{NGP}} \cos b \cos(\ell_{\text{NCP}} - \ell) \quad (24.19)$$

$$\cos \delta \sin(\alpha - \alpha_{\text{NGP}}) = \cos b \sin(\ell_{\text{NCP}} - \ell) \quad (24.20)$$

$$\cos \delta \cos(\alpha - \alpha_{\text{NGP}}) = \cos \delta_{\text{NGP}} \sin b - \sin \delta_{\text{NGP}} \cos b \cos(\ell_{\text{NCP}} - \ell). \quad (24.21)$$

Note that care must be taken in inverting these expressions to solve for α or ℓ , because computers and electronic calculators may return the wrong (default) quadrant when the calculation of an inverse trigonometric function is performed. A graphical form of the transformations that does not suffer from this multiplicity is presented in Fig. 24.18.

A Cylindrical Coordinate System for Galactic Motions

The motions of stars in the solar neighborhood allow us to glean important clues regarding the large-scale structure of the Galaxy. Although the Galactic coordinate system is useful for representing the locations of objects within the Galaxy as seen from Earth, it is not the most convenient choice for studying kinematics and dynamics. One reason is that the Sun, which is the origin of the Galactic system, is itself moving about the center of the Galaxy. In addition, a coordinate system centered on the Sun constitutes a noninertial reference frame with respect to Galactic motions.

Therefore, to complement the Galactic coordinate system, a cylindrical coordinate system is used that places the center of the Galaxy at the origin. In this system the radial coordinate R increases outward, the angular coordinate θ is pointed in the direction of rotation of the Galaxy, and the vertical coordinate z increases to the north (see Fig. 24.19). Furthermore, the corresponding velocity components are traditionally labeled

$$\Pi \equiv \frac{dR}{dt}, \quad \Theta \equiv R \frac{d\theta}{dt}, \quad Z \equiv \frac{dz}{dt}. \quad (24.22)$$

It is worth noting that this set of directional choices results in a left-handed coordinate system instead of a more conventional right-handed one. This occurs because, when viewed from the north Galactic pole, the Galaxy rotates clockwise, rather than counterclockwise. (Fortunately, we are not going to be concerned with cross products in this analysis. Recall that cross products are right-handed by definition.)

Peculiar Motions and the Local Standard of Rest

Since all of our observations are made from Earth (or at least relatively near it, in the case of satellites), and because we can transform those observations to Sun-centered ones by removing any effects resulting from the rotational and orbital motions of Earth or motions of the spacecraft, we will consider the Sun as the site of all observations of the Galaxy. In particular, since the Earth–Sun distance is very small when compared to any distances on a Galactic scale, we need not concern ourselves with the change in position, only with changes in velocity.

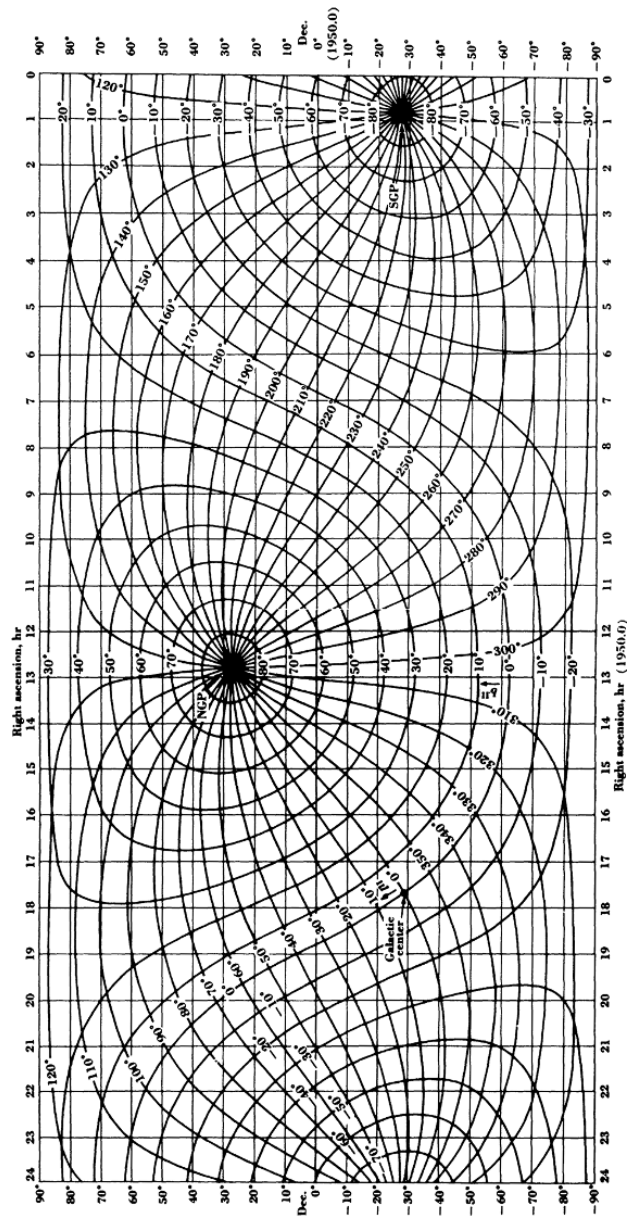


FIGURE 24.18 A chart for converting between equatorial and Galactic coordinates. (From Kraus, *Radio Astronomy*, Second Edition, Cygnus-Quasar Books, 1986, used by permission.)

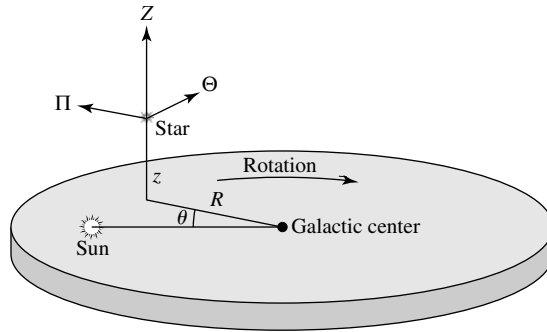


FIGURE 24.19 The cylindrical coordinate system used to analyze Galactic kinematics. (Note that some authors define the radial coordinate to be positive toward the Galactic center. It should also be noted that the coordinate system defined here is *not* a right-handed coordinate system.)

Before we can hope to make a final transformation from the Sun to the center of the Galaxy, however, we must also understand the Sun's motion. In fact, the Sun does not follow a simple planar orbit or even a closed nonplanar orbit; instead, as it moves about the Galactic center, it is also currently moving slowly inward (in the negative- R direction) and farther north, away from the midplane (positive- z direction).

To investigate the motion of the Sun and the other stars in the solar neighborhood, we will first define the **dynamical local standard of rest** (dynamical LSR) to be a point that is *instantaneously* centered on the Sun and moving in a *perfectly circular orbit* along the solar circle about the Galactic center. An alternative definition for the LSR known as the **kinematic local standard of rest** is based on the average motions of stars in the solar neighborhood. Although with the right choice of reference stars, dynamical and kinematic LSRs agree quite well, it can be shown that the kinematic LSR systematically lags behind the dynamical LSR. Throughout the remainder of this text, “the LSR” will always refer to the dynamical LSR.

The velocity components of the LSR must be

$$\Pi_{\text{LSR}} \equiv 0, \quad \Theta_{\text{LSR}} \equiv \Theta_0, \quad Z_{\text{LSR}} \equiv 0,$$

where $\Theta_0 \equiv \Theta(R_0)$ and R_0 is the solar Galactocentric distance. Note that once the LSR is chosen, the Sun immediately begins to drift away from it, implying that we would effectively need to redefine the reference point constantly. In reality this is not a significant problem because (fortunately) the 230-Myr orbital period of the LSR is very long compared to the time since modern telescopic observations began (and even longer compared to the lifetime of a typical research grant); consequently, there has not been sufficient time for the effect to become noticeable.

The velocity of a star *relative to the LSR* is known as the star's **peculiar velocity** and is given by

$$\mathbf{V} = (V_R, V_\theta, V_z) \equiv (u, v, w), \quad (24.23)$$

where

$$u = \Pi - \Pi_{\text{LSR}} = \Pi, \quad (24.24)$$

$$v = \Theta - \Theta_{\text{LSR}} = \Theta - \Theta_0, \quad (24.25)$$

$$w = Z - Z_{\text{LSR}} = Z. \quad (24.26)$$

The Sun's peculiar velocity relative to the LSR is generally referred to simply as the **solar motion**.

The average of u and w for all stars in the solar neighborhood, excluding the Sun, should be nearly zero if we assume an axisymmetric Galaxy. The reason is that with symmetries about both the rotation axis and the midplane, there should be as many stars moving inward as outward, and there should be as many stars moving toward the north Galactic pole as toward the south Galactic pole. In reality this is not quite true because the Galaxy is not precisely axisymmetric, but the error is not significant for our purposes here. As a result, we shall assume that, summing over a sample of N nearby stars,

$$\langle u \rangle = \frac{1}{N} \sum_{i=1}^N u_i \simeq 0, \quad (24.27)$$

$$\langle w \rangle = \frac{1}{N} \sum_{i=1}^N w_i \simeq 0. \quad (24.28)$$

The same assumption cannot be made for the v component, however. To see why, consider the situations depicted in Fig. 24.20. Stars with different average orbital radii must follow paths that bring them very close to the LSR if they are to be considered members of the solar neighborhood and eligible for inclusion in the calculation of $\langle v \rangle$. If we consider the special case that $u = 0$ and $w = 0$ for all stars in our sample, then the stars must be at either their most distant point from the Galactic center (**apogalacticon**) or their closest approach to the Galactic center (**perigalacticon**) when they coincide with the LSR; this is true of A and B , respectively. Then, for the two stars to follow their specified orbits, it is necessary

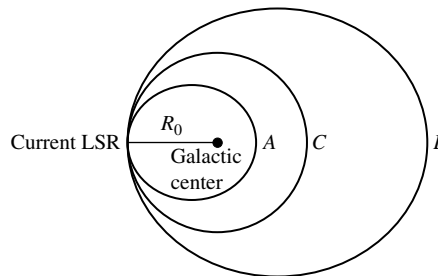


FIGURE 24.20 The orbits of three hypothetical stars intersecting at the LSR. A and B represent stars in elliptical orbits with semimajor axes $a_A < R_0$ and $a_B > R_0$, respectively. C represents a star following the perfectly circular path of the LSR.

that $\Theta_A(R_0) < \Theta_0$ and $\Theta_B(R_0) > \Theta_0$. This implies that $v_A < 0$ and $v_B > 0$. Finally, since more stars reside inside the solar Galactocentric distance than beyond it (see, for example, Eq. 24.9),

$$\langle v \rangle < 0. \quad (24.29)$$

(This explains why the kinematic LSR systematically lags behind the dynamical LSR, as mentioned previously.)

The velocity that is measured for a particular star relative to the Sun is just the difference between the star's peculiar velocity and the solar motion with respect to the LSR, or

$$\Delta u \equiv u - u_\odot, \quad \Delta v \equiv v - v_\odot, \quad \Delta w \equiv w - w_\odot.$$

Using the average values of the stellar peculiar velocity components given by Eqs. (24.27), (24.28), and (24.29), and solving for the solar motion, we have

$$u_\odot = -\langle \Delta u \rangle, \quad (24.30)$$

$$v_\odot = \langle v \rangle - \langle \Delta v \rangle, \quad (24.31)$$

$$w_\odot = -\langle \Delta w \rangle. \quad (24.32)$$

The u and w components of the solar motion simply reflect the averaged *relative* velocities of the other stars with respect to the Sun in the R and z directions. Qualitatively, these stars appear to be “streaming” past the Sun as it moves through space (see Problem 24.13).

To find the v component of the solar motion, we must first determine the average value of v for stars in the solar neighborhood. Unfortunately, this requires a theory of Galactic motion that is beyond the scope of the present argument. However, qualitatively the procedure involves deriving an analytical expression for $\langle v \rangle$ in terms of the radial variation in the number density of stars in the solar neighborhood. The justification for this relationship lies in the argument made concerning the lag of stellar motions behind the LSR due to the increase in number density with decreasing Galactocentric distance. The result is an equation of the form

$$\langle v \rangle = C\sigma_u^2,$$

where C is a constant and

$$\sigma_u \equiv \langle u^2 \rangle^{1/2}$$

measures the spread in the R components of the peculiar velocities of stars in the solar neighborhood with respect to the LSR; of course, $\sigma_u^2 = \langle \Pi^2 \rangle$.

Note that σ_u is related to the standard deviation of the velocity distribution, defined as

$$\text{standard deviation} \equiv \frac{1}{\sqrt{N}} \left[\sum_{i=1}^N (u - \langle u \rangle)^2 \right]^{1/2}.$$

In the special case that $\langle u \rangle = 0$, then $\sigma_u = \langle u^2 \rangle^{1/2}$ is identical to the standard deviation. σ_u is known as the **velocity dispersion** in u .

Turning again to Fig. 24.20, we can see why σ_u should correlate with $\langle v \rangle$. A stellar sample that produces a larger dispersion in u means that a wider range of elliptical orbits are included. This results in a more negative average value of v for the sample because of the larger fraction of the stellar population with $R < R_0$; there are more stars in the sample with $v < 0$ than with $v > 0$. Alternatively, as σ_u^2 decreases, fewer stars will have orbits that are appreciably noncircular and $\langle v \rangle$ will approach zero. From Eq. (24.31),

$$\langle \Delta v \rangle = C\sigma_u^2 - v_\odot$$

and $-v_\odot$ is simply the ordinate intercept on a graph of $\langle \Delta v \rangle$ versus σ_u^2 ; see Problem 24.14. C is the slope of the linear relationship.

The components of the Sun's peculiar velocity are

$$u_\odot = -10.0 \pm 0.4 \text{ km s}^{-1}, \quad (24.33)$$

$$v_\odot = 5.2 \pm 0.6 \text{ km s}^{-1}, \quad (24.34)$$

$$w_\odot = 7.2 \pm 0.4 \text{ km s}^{-1}, \quad (24.35)$$

so that relative to the LSR, the Sun is moving (a) toward the Galactic center, (b) more rapidly in the direction of Galactic rotation, and (c) north out of the Galactic plane. Overall, the solar motion is approximately 13.4 km s^{-1} toward a point in the constellation of Hercules. The point toward which the Sun is moving is called the **solar apex**; the point away from which the Sun is retreating is the **solar antapex** (located in Columba). It is important to note, however, that the exact value for the solar motion and the location of the solar apex depend on the choice of reference stars.

Now that the solar motion is known, the velocities of stars relative to the Sun can be transformed into peculiar motions relative to the LSR. It is then possible to plot one component of peculiar motion against another for a specified sample of stars in the solar neighborhood in order to obtain important information about their kinematics. Such plots result in patterns known as **velocity ellipsoids**. As depicted in a diagram of u versus v (Fig. 24.21), when young, metal-rich main-sequence A stars are used, the range in velocities about the LSR is fairly limited (a small dispersion), for older K giants a wider variation in both u and v is observed, and when old, metal-poor red dwarfs are plotted, the spread is even larger (a large dispersion). The same general behavior is seen in plots of w versus v , whereas a much more symmetric diagram results when w is plotted against u .

Specific features of Fig. 24.21 are worth discussing in more detail. First is the very noticeable relationship between metallicity and velocity dispersion, called a **velocity–metallicity relation**. When the velocity–metallicity and age–metallicity relations are combined, the velocity ellipsoids suggest that the oldest stars in the Galaxy have the widest range of peculiar velocities, a trend that is evident in all three coordinates. Because stars with the smallest peculiar velocities do not drift away from the LSR as quickly, they must occupy orbits that are similar to that of the LSR, implying that these young stars are members of the thin disk. On the other hand, the stars with the largest peculiar velocities follow very different paths about the center of the Galaxy. In particular, stars with large $|w|$ must be passing through

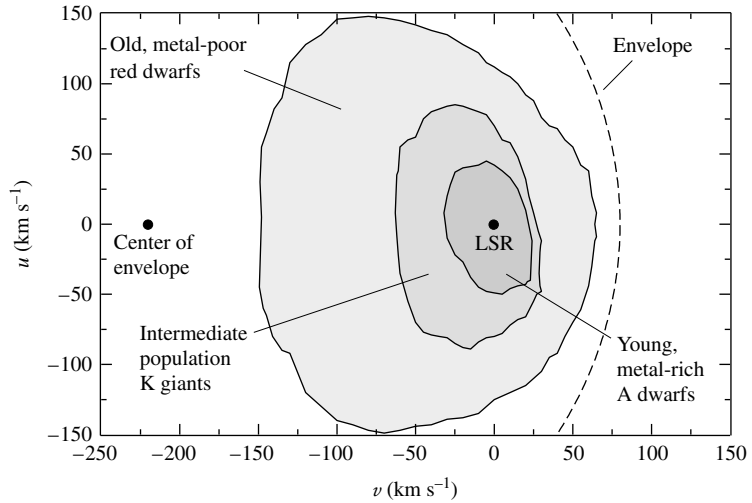


FIGURE 24.21 A schematic diagram of the peculiar velocity components v and u for stars in the solar neighborhood. The innermost contour represents metal-rich main-sequence A stars, the middle contour depicts older K giants, and the outer contour indicates very metal-poor red dwarfs. The LSR is located at $(v, u) = (0, 0)$. An enveloping circle with an approximate radius of 300 km s^{-1} and centered at $v = -220 \text{ km s}^{-1}$ reveals the orbital velocity of the LSR.

the solar neighborhood on trajectories that will carry them to great distances above and below the disk. These old, metal-poor stars are the high-velocity stars that were mentioned in the last section as being members of the stellar halo.

A second common feature of peculiar velocity diagrams is the clear asymmetries in the velocity ellipsoids along the v -axis as a function of metallicity or age (this effect is known as **asymmetric drift**). Few stars are observed with $v > +65 \text{ km s}^{-1}$, but there are metal-poor RR Lyraes and subdwarfs with $v < -250 \text{ km s}^{-1}$. In fact, as suggested by the dashed line in Fig. 24.21, a nearly circular “envelope” with a radius of roughly 300 km s^{-1} can be drawn around the high-velocity stars. The center of the velocity envelope appears to be near $v = -220 \text{ km s}^{-1}$ for both u - v and w - v diagrams. If on average the stellar halo is rotating very slowly (if at all), then the orbital velocity of the LSR should reveal itself as a point of symmetry along the v -axis. This is because halo stars with $\Theta \simeq 0$ (no velocity component in the direction of Galactic rotation) should exhibit peculiar v velocities that simply reflect the motion of the LSR (i.e., $v \simeq -\Theta_0$). Stars that have orbital components that are in the opposite sense from the overall Galactic rotation direction have $v < -\Theta_0$. By this argument it appears that the orbital speed of the LSR is

$$\Theta_0(R_0) = 220 \text{ km s}^{-1}, \quad (24.36)$$

the presently accepted IAU standard. This value has also been measured using groups of external galaxies for reference.⁸

⁸Kuijken and Tremaine (1994) have suggested that the IAU value of Θ_0 may be too large. Based on a set of self-consistent solutions to various Galactic parameters, they argue for $\Theta_0 = 180 \text{ km s}^{-1}$.

Example 24.3.1. An estimate of the mass of the Milky Way Galaxy interior to the solar Galactocentric distance can be made using Kepler's third law, together with R_0 and Θ_0 . Using $R_0 = 8$ kpc and $\Theta_0 = 220 \text{ km s}^{-1}$, the orbital period of the LSR is

$$P_{\text{LSR}} = \frac{2\pi R_0}{\Theta_0} = 230 \text{ Myr.}$$

Assuming that the mass of the Galaxy within the solar circle is much greater than the mass of a test particle orbiting along with the LSR, and that the bulk of the Galaxy's mass is distributed spherically symmetrically, Kepler's third law (Eq. 2.37) gives

$$M_{\text{LSR}} = \frac{4\pi^2 R_0^3}{G P_{\text{LSR}}^2} = 8.8 \times 10^{10} M_{\odot}.$$

This value compares well with the estimate of the mass of luminous matter quoted on page 896, but it is much less than the total mass estimate of the Galaxy when the dark matter halo is included.

In 1927, Jan Oort (1900–1992) proposed that since no stars had been observed with $v > +65 \text{ km s}^{-1}$, the escape velocity of the Galaxy must be $\Theta_0 + 65 \text{ km s}^{-1} \sim 300 \text{ km s}^{-1}$ relative to the Galactic center. In fact, we know today that a small number of extremely high-velocity stars do exist in the solar neighborhood with speeds of $\sim 500 \text{ km s}^{-1}$ relative to the center of the Galaxy. Since these stars have not escaped from the Galaxy, it seems that the strong asymmetry near $v \sim +65 \text{ km s}^{-1}$ simply points to a deficiency in very high-velocity stars; see Problem 24.15.

Differential Galactic Rotation and Oort's Constants

In 1927 Jan Oort also derived a series of relations that have become the framework with which astronomers have attempted to determine the differential rotation curve of the Galactic disk. To simplify the discussion, we will assume that all motions are circular about the center of the Galaxy.

Consider the situation depicted in Fig. 24.22. Assume that the Sun (at O) and a star or some other object (at S) are orbiting the Galactic center (at C) in the Galactic midplane. The velocity vector that is measured between the Sun and the star at point S is the *relative velocity* between the two objects. Therefore, in order to compare the observed velocity vector to the object's true velocity with respect to the Galactic center, it is necessary to consider the difference between the star's motion and the Sun's motion. Of course, in practice it is not the relative space motion that is measured directly but the radial velocity and the proper motion, with the proper motion being converted into the transverse velocity if the distance d to the star is known.

If the line of sight is in the direction of the Galactic longitude ℓ , and if $\Theta(R)$ is the orbital velocity curve as a function of distance from the Galactic center, then the relative radial and transverse velocities of the star are, respectively,

$$v_r = \Theta \cos \alpha - \Theta_0 \sin \ell,$$

$$v_t = \Theta \sin \alpha - \Theta_0 \cos \ell,$$

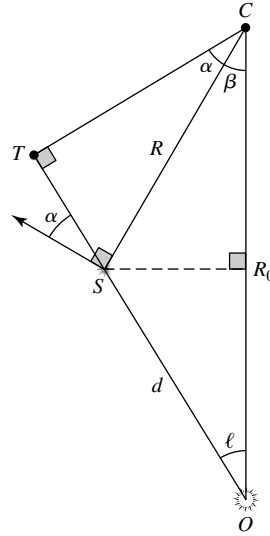


FIGURE 24.22 The geometry of analyzing differential rotation in the Galactic plane. The Sun is at point O , the center of the Galaxy is located at C , and the star is at S , located a distance d from the Sun. ℓ is the Galactic longitude of the star at S , and α and β are auxiliary angles. The directions of motion reflect the clockwise rotation of the Galaxy as viewed from the NGP.

where Θ_0 is the orbital velocity of the Sun in the idealized case of perfectly circular motion (actually the orbital velocity of the LSR) and α is defined in the figure. Defining the *angular-velocity* curve to be

$$\Omega(R) \equiv \frac{\Theta(R)}{R},$$

the relative radial and transverse velocities become

$$v_r = \Omega R \cos \alpha - \Omega_0 R_0 \sin \ell,$$

$$v_t = \Omega R \sin \alpha - \Omega_0 R_0 \cos \ell.$$

Now, by referring to the geometry of Fig. 24.22 and considering the right triangle $\triangle OTC$, we find

$$R \cos \alpha = R_0 \sin \ell,$$

$$R \sin \alpha = R_0 \cos \ell - d.$$

Substituting these relations into the previous expressions, we have

$$v_r = (\Omega - \Omega_0) R_0 \sin \ell, \quad (24.37)$$

$$v_t = (\Omega - \Omega_0) R_0 \cos \ell - \Omega d. \quad (24.38)$$

Equations (24.37) and (24.38) are valid as long as the assumption of circular motion is justified.

Although the Sun's motion around the Galactic center is not perfectly circular, its peculiar velocity relative to the LSR is small compared to Θ_0 (only about 6%). As a result, to a first approximation, Eqs. (24.37) and (24.38) provide a reasonable estimate of $\Omega = \Omega(R)$ if the other parameters are known. However, d is in general very difficult to measure unless the object is close enough to yield to trigonometric parallax, or perhaps some other reasonably reliable distance estimate may be applicable (for instance, the star may be a Cepheid variable).

Another complication arises because of the effects of interstellar extinction. Our ability to observe Galactic structure to great distances is severely limited at visual wavelengths. Unless we are making observations in relatively unobscured directions such as Baade's window, we are restricted to seeing stars out to a few thousand parsecs from the Sun. One important exception to this constraint (to be discussed in more detail below) is the 21-cm-wavelength band of H I; virtually the entire Galaxy is optically thin to 21-cm radiation, making that wavelength band a valuable tool for studying Galactic structure.

Because of the distance limitation at optical wavelengths, Oort derived a set of approximate equations for v_r and v_t that are valid only in the region near the Sun. Despite this restriction, these alternative formulae are still able to provide a surprising amount of information about the large-scale structure of the Galaxy.

We make the assumption here that $\Omega(R)$ is a smoothly varying function of R so that the Taylor expansion of $\Omega(R)$ about $\Omega_0(R_0)$ is given by

$$\Omega(R) = \Omega_0(R_0) + \left. \frac{d\Omega}{dR} \right|_{R_0} (R - R_0) + \cdots$$

Thus, to first order, the *difference* between Ω and Ω_0 is

$$\Omega - \Omega_0 \simeq \left. \frac{d\Omega}{dR} \right|_{R_0} (R - R_0),$$

and the *approximate value* of Ω is

$$\Omega \simeq \Omega_0.$$

If we also make use of the identity $\Omega = \Theta/R$, Eqs. (24.37) and (24.38) become (after some rearrangement)

$$\begin{aligned} v_r &\simeq \left[\left. \frac{d\Theta}{dR} \right|_{R_0} - \frac{\Theta_0}{R_0} \right] (R - R_0) \sin \ell, \\ v_t &\simeq \left[\left. \frac{d\Theta}{dR} \right|_{R_0} - \frac{\Theta_0}{R_0} \right] (R - R_0) \cos \ell - \Omega_0 d. \end{aligned}$$

From Fig. 24.22 it is clear that

$$R_0 = d \cos \ell + R \cos \beta \simeq d \cos \ell + R,$$

the latter result being due to the small-angle approximation $\cos \beta \simeq 1$, since $d \ll R_0$ implies that $\beta \ll 1$ radian. Finally, using the appropriate trigonometric identities and defining the **Oort constants**

$$A \equiv -\frac{1}{2} \left[\left. \frac{d\Theta}{dR} \right|_{R_0} - \frac{\Theta_0}{R_0} \right], \quad (24.39)$$

$$B \equiv -\frac{1}{2} \left[\left. \frac{d\Theta}{dR} \right|_{R_0} + \frac{\Theta_0}{R_0} \right], \quad (24.40)$$

we have

$$v_r \simeq Ad \sin 2\ell, \quad (24.41)$$

$$v_t \simeq Ad \cos 2\ell + Bd. \quad (24.42)$$

To understand the functional dependence of the Oort formulae on Galactic longitude, consider the orbits of nearby stars shown in Fig. 24.23(a). For stars in the directions $\ell = 0^\circ$ and $\ell = 180^\circ$, the lines of sight are perpendicular to their motions relative to the LSR. As a result, the radial velocity must be zero. For $\ell = 90^\circ$ or 270° , the stars being observed are in essentially the same circular orbit as the Sun and are moving with the same speed, so again $v_r = 0 \text{ km s}^{-1}$. At intermediate angles the situation is somewhat more complicated, however. For instance, if we assume that in the neighborhood of the Sun, $\Omega(R)$ is monotonically decreasing outward, then at $\ell = 45^\circ$, the star being observed is closer to the Galactic center and is “outrunning” the Sun; hence a positive radial velocity would be measured. For $\ell = 135^\circ$, the Sun is “overtaking” the star and a negative radial velocity results. At $\ell = 225^\circ$, the Sun is moving away from the star, producing a positive radial velocity, and at $\ell = 315^\circ$, the star is gaining on the Sun, causing the observed radial velocity to be negative. The overall result is the double sine function shown in Fig. 24.23(b). A similar analysis shows that the transverse-velocity curve is a double cosine function plus an additive constant; see Problem 24.17. For a sample of stars, all with similar distances d , the amplitudes of the v_r and v_t curves (Eqs. 24.41 and 24.42, respectively) give A , and the vertical offset in v_t gives B .

It is now possible to derive several important relationships between the Oort constants A and B , and the *local* parameters of Galactic rotation, R_0 , Θ_0 , $\Omega_0 = \Theta_0/R_0$, and $(d\Theta/dR)_{R_0}$. For instance, from Eqs. (24.39) and (24.40), we immediately find that

$$\Omega_0 = A - B, \quad (24.43)$$

$$\left. \frac{d\Theta}{dR} \right|_{R_0} = -(A + B). \quad (24.44)$$

Yet another useful relation can be found by considering the largest radial velocity seen along the line of sight at a constant Galactic longitude ℓ . Turning again to Fig. 24.22, the star with the maximum observable radial velocity will be located at point T , the position

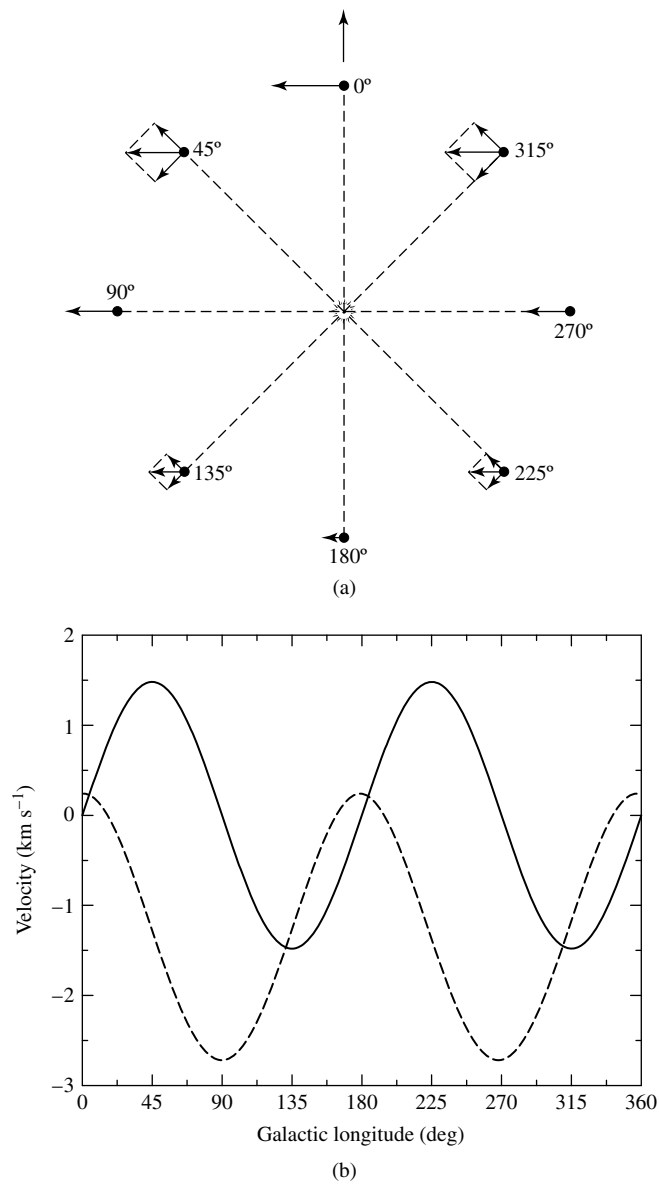


FIGURE 24.23 (a) The differential rotation of stars near the Sun is revealed through the dependence of radial and transverse velocities on Galactic longitude. (b) Radial velocity is proportional to $\sin 2\ell$ (solid line), and transverse velocity is a function of $\cos 2\ell$ (dashed line). The curves depict stars located 100 pc from the Sun with $A = 14.8 \text{ km s}^{-1} \text{ kpc}^{-1}$ and $B = -12.4 \text{ km s}^{-1} \text{ kpc}^{-1}$.

where $\alpha = 0^\circ$. It is at this *tangent point* that the distance to the center of the Galaxy will be a minimum and $\Theta(R)$ will be a maximum (if we can assume that $\Theta(R)$ is monotonically decreasing from the center outward). Moreover, the orbital-velocity vector is directed along the line of sight at that position. This minimum distance from the center of the Galaxy is given by

$$R_{\min} = R_0 |\sin \ell| = \pm R_0 \sin \ell \quad \begin{cases} +, \text{ if } 0^\circ \leq \ell \leq 90^\circ \\ -, \text{ if } 270^\circ \leq \ell \leq 360^\circ \end{cases}$$

and the maximum radial velocity is

$$v_{r,\max} = \pm \Theta(R_{\min}) - \Theta_0(R_0) \sin \ell \quad \begin{cases} +, \text{ if } 0^\circ \leq \ell \leq 90^\circ \\ -, \text{ if } 270^\circ \leq \ell \leq 360^\circ. \end{cases}$$

If we now restrict our observations to Galactic longitudes near but less than 90° , or near but greater than 270° (i.e., inside the solar circle), then $d \ll R_0$, $R \sim R_0$, and $\Theta(R)$ can be expressed in terms of a Taylor expansion about Θ_0 :

$$\Theta(R_{\min}) = \Theta_0(R_0) + \left. \frac{d\Theta}{dR} \right|_{R_0} (R_{\min} - R_0) + \dots$$

Substituting into the expression for $v_{r,\max}$, retaining first-order terms, and making use of Eq. (24.39), we find

$$v_{r,\max} \simeq 2AR_0 (\pm 1 - \sin \ell) \quad \begin{cases} +, \text{ if } 0^\circ \leq \ell \leq 90^\circ \\ -, \text{ if } 270^\circ \leq \ell \leq 360^\circ. \end{cases} \quad (24.45)$$

One last relation, which we will not attempt to derive here but merely include for completeness, associates A and B with the dispersions of peculiar velocities in the R and θ directions:

$$\frac{-B}{A - B} = \frac{\sigma_v^2}{\sigma_u^2}. \quad (24.46)$$

Equations (24.43–24.46) place additional constraints on the values of R_0 and Θ_0 beyond the direct observations discussed above. In fact, because A and B provide critical information about Galactic differential rotation in the solar neighborhood, considerable effort has gone into determining these constants. Based on results from the Hipparcos astrometry mission, values of

$$A = 14.8 \pm 0.8 \text{ km s}^{-1} \text{ kpc}^{-1} \quad (24.47)$$

$$B = -12.4 \pm 0.6 \text{ km s}^{-1} \text{ kpc}^{-1} \quad (24.48)$$

appear to be consistent with the available data, although there remains debate over the best choices for A and B .

Hydrogen 21-cm Line as a Probe of Galactic Structure

To determine the large-scale velocity structure of the Galactic disk, we must return to the more general expressions for v_r and v_t that do not rely on first-order Taylor series expansions.

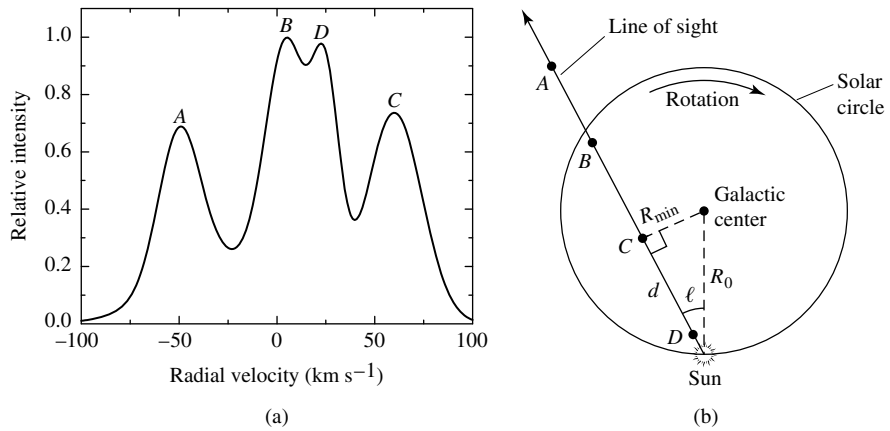


FIGURE 24.24 (a) A typical 21-cm H I line profile. (b) The line profile is produced by observing several gas clouds along a particular line of sight. Because of differential Galactic rotation, each cloud has a different radial velocity relative to the Sun.

As has already been mentioned, 21-cm emission from H I is able to penetrate virtually the entire Galaxy, making it an indispensable tool in probing the structure of the Milky Way. By measuring v_r as a function of ℓ , the Galactic rotation curve can be determined from Eq. (24.37), provided that the distance of the emitting region from the Sun can be found.

Figure 24.24(a) shows a typical intensity distribution for the 21-cm emission line of H I along a particular line of sight. When a specific cloud is encountered along the line of sight, the wavelength of the radiation from that cloud is Doppler shifted because of the effects of differential Galactic rotation. Furthermore, the intensity of the radiation at a given wavelength (or velocity) is proportional to the number of hydrogen atoms along the line of sight in the cloud. The peaks of the line profile shown in Fig. 24.24(a) correspond to the clouds shown in Fig. 24.24(b).

The principal difficulty in using 21-cm radiation to determine $\Omega(R)$, and hence $\Theta(R)$, lies in measuring d . This problem can be overcome by selecting the largest radial velocity measured along each line of sight, which must originate in the region $R_{\min}$ from the Galactic center, implying that $d = R_0 \cos \ell$. By measuring $v_{r,\max}$ for $0^\circ < \ell < 90^\circ$ and $270^\circ < \ell < 360^\circ$, we can determine the rotation curve within the solar Galactocentric radius. Unfortunately, this technique does not work for Galactic longitudes $90^\circ < \ell < 270^\circ$ because there is no unique orbit for which a maximum radial velocity can be observed. The method also tends to break down near $\ell = 90^\circ$ and $\ell = 270^\circ$ because v_r becomes rather insensitive to changes in distance from the Sun. For longitudes within approximately 20° of the Galactic center, further problems develop; clouds that have markedly noncircular motions exist in that region, perhaps because of the gravitational perturbations of the central bar, so the assumptions underlying the preceding analysis are not valid.

The Flat Rotation Curve and Evidence of Dark Matter

To measure $\Theta(R)$ for $R > R_0$, we must rely on objects available in the Galactic plane, such as Cepheids, for which we can directly obtain distances. These data suggest that the

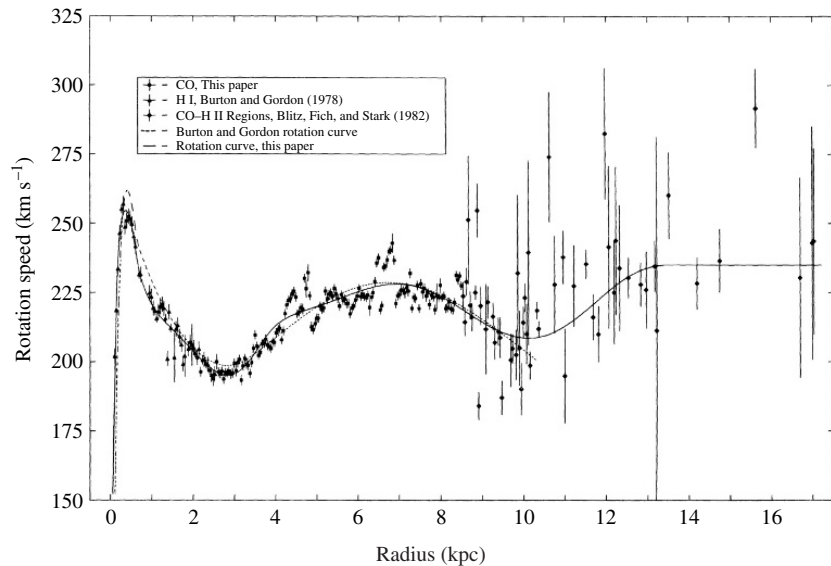


FIGURE 24.25 The rotation curve of the Milky Way Galaxy. The 1985 IAU standard values of $R_0 = 8.5$ kpc and $\Theta_0 = 220 \text{ km s}^{-1}$ have been assumed. (Figure adapted from Clemens, *Ap. J.*, 295, 422, 1985.)

rotation curve of the Galaxy does not decrease significantly with distance beyond R_0 and may actually increase somewhat (meaning that for the Oort constants, $A < -B$ near R_0). Combining all of the available data, a possible form for the rotation curve of the Galaxy is shown in Fig. 24.25.

It came as a great surprise to astronomers to discover that the Galactic rotation curve is essentially constant beyond R_0 . According to Newtonian mechanics, if most of the mass were interior to the solar circle, the rotation curve should drop off as $\Theta \propto R^{-1/2}$, a behavior referred to as Keplerian motion (see Problem 24.18). The fact that it does not implies that a significant amount of mass exists in the Galaxy beyond R_0 . This result was particularly unexpected since most of the luminosity in the Galaxy is produced by matter residing inside the solar Galactocentric radius.

The data for the Milky Way are supported by observations of other spiral galaxies, such as those obtained by Vera Rubin and her collaborators in the late 1970s. Figure 24.26 shows a spectrograph slit superimposed on NGC 2998, a galaxy in the constellation of Ursa Major, 96 Mpc from Earth. Below that image is a portion of the spectrum in a wavelength region near $H\alpha$. The left side of the slit recorded blueshifted light, and the light on the right side was redshifted. The Doppler shifts were then translated into radial velocities, and a corresponding rotation curve was determined.

Similar rotation curves have also been measured for a number of other spiral galaxies (see Fig. 24.27). With the exception of the innermost regions (to be discussed in Section 24.4 for the Milky Way), there is a rapid rise in rotation speed with distance out to a few kiloparsecs from the center. This type of rotation is referred to as **rigid-body rotation** because when $\Theta \propto R$, $\Omega = \Theta/R$ is a constant and all stars have the same orbital period about the Galactic

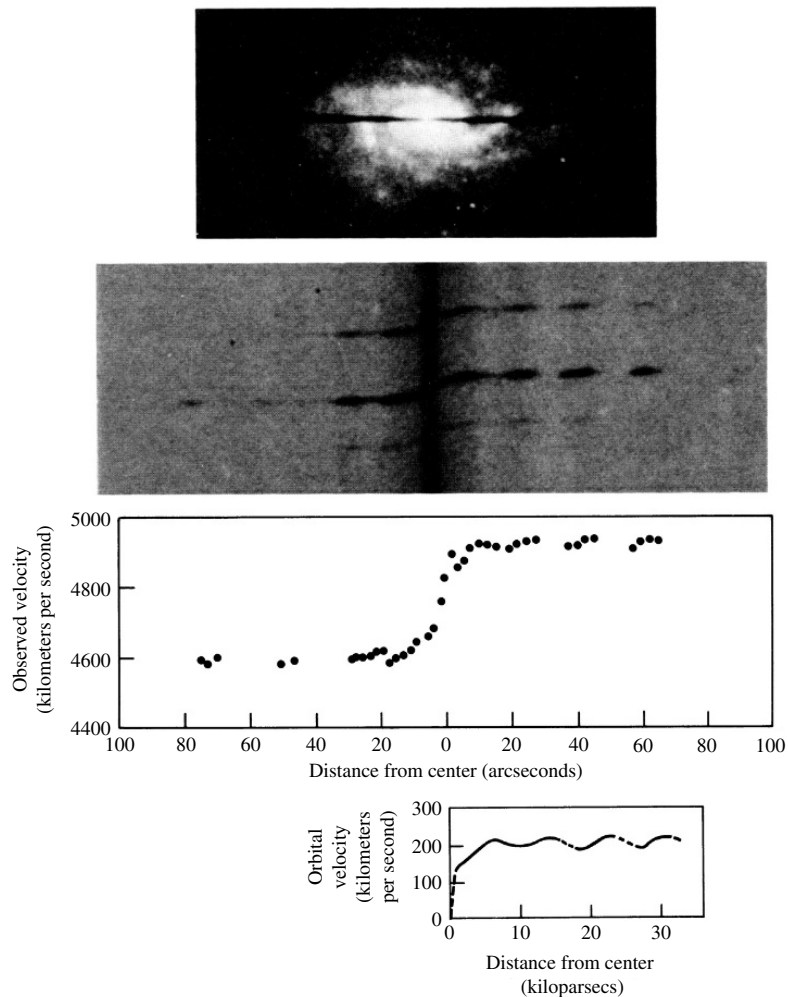


FIGURE 24.26 The rotation curve of NGC 2998 was measured using a slit spectrograph. The H α wavelength region is shown. Note that overall, the entire galaxy is receding from us at a speed of 4800 km s⁻¹. (Adapted from a figure courtesy of Vera Rubin, 1983.)

center, just as a rigid object would. Beyond a few kiloparsecs, nearly **flat rotation curves** continue out to the edge of the measurements.

Since Galactic rotation depends on the distribution of mass, a great deal can be learned about the matter in galaxies by studying these curves. For instance, rigid-body rotation near the Galactic center implies that the mass must be roughly spherically distributed and the density nearly constant (see Problem 24.20). On the other hand, flat rotation curves suggest that the bulk of the mass in the outer portions of the Galaxy are spherically distributed with a density law that is proportional to r^{-2} .

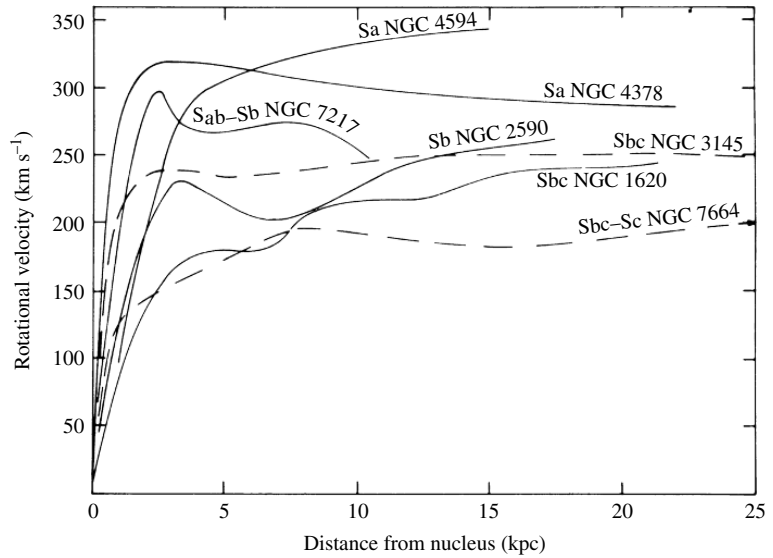


FIGURE 24.27 A series of rotation curves for spiral galaxies. (Figure adapted from Rubin, Ford, and Thonnard, *Ap. J. Lett.*, 225, L107, 1978.)

To see this, assume that $\Theta(r) = V$, where V is a constant.⁹ Then, from the equation for centripetal force and Newton's law of gravity, the force acting on a star of mass m due to the mass M_r of the Galaxy interior to the star's position at r is

$$\frac{mV^2}{r} = \frac{GM_r m}{r^2},$$

if spherical symmetry is assumed. Solving for M_r ,

$$M_r = \frac{V^2 r}{G}, \quad (24.49)$$

and differentiating with respect to the radius of the distribution,

$$\frac{dM_r}{dr} = \frac{V^2}{G}.$$

If we now borrow the equation for mass conservation in a spherically symmetric system from stellar structure theory, Eq. (10.7),

$$\frac{dM_r}{dr} = 4\pi r^2 \rho,$$

⁹We are using r for a spherically symmetric mass distribution here, rather than R for cylindrical rotation in the Galactic plane. However, to obtain a rotation curve within the Galactic plane, we need only consider the special case of $r = R$.

we see that the mass density in the outer regions of the Galaxy must vary as

$$\rho(r) = \frac{V^2}{4\pi Gr^2}. \quad (24.50)$$

This r^{-2} density dependence is very different from the form determined by star counts in the portion of the Galaxy beyond the solar Galactocentric radius. Recall from Eq. (24.14) that the number density of stars in the luminous stellar halo is believed to vary as $r^{-3.5}$, a much more rapid drop-off than is evident from the flat rotation curve. It was this discrepancy that so surprised astronomers. As we mentioned at the end of Section 24.2, it appears that the majority of the mass in the Galaxy is in the form of nonluminous (dark) matter. Only through its gravitational influence on the luminous component of our Galaxy and satellite galaxies like the LMC and the SMC, and through gravitational lensing of light from background sources, does the dark matter make its presence known.

One modification to Eq. (24.50) that has been made by many researchers is to force the density function to approach a constant value near the center, rather than diverge. Such a model is also consistent with the observational evidence of rigid-body rotation. As a result, one commonly used density profile for the Milky Way's dark matter halo is assumed to be of the form

$$\rho(r) = \frac{\rho_0}{1 + (r/a)^2}, \quad (24.51)$$

where ρ_0 and a are chosen as parametric fits to the overall rotation curve. Note that for $r \gg a$, the r^{-2} dependence is obtained, and $\rho \sim \text{constant}$ when $r \ll a$. A similar profile is often used for modeling other galaxies as well, with different choices for ρ_0 and a .

It is important to point out that Eq. (24.51) cannot be correct to arbitrarily large values of r . The reason for this is that the total amount of mass in the Galaxy would increase without bound since $M_r \propto r$. As a result, the density function for the dark matter halo must eventually terminate or at least decrease sufficiently rapidly that the mass integral $\int_0^\infty \rho(r) 4\pi r^2 dr$ remains finite.

An alternative form of the dark matter halo density distribution was proposed by Julio Navarro, Carlos Frenk, and Simon White in 1996. Using a commonly assumed form of dark matter dynamics known as **cold dark matter** (CDM), Navarro, Frenk, and White ran numerical simulations of the formation of dark matter halos over a wide range of size and mass scales, ranging from dwarf galaxies to rich clusters of galaxies. Their simulations revealed that a “universal” profile of the form

$$\rho_{\text{NFW}}(r) = \frac{\rho_0}{(r/a)(1 + r/a)^2} \quad (24.52)$$

was applicable over an enormous range with appropriate choices of ρ_0 and a (this is the profile first given by Eq. 24.15). The NFW density profile behaves approximately like a $1/r^2$ profile over much of the halo but is shallower ($\sim 1/r$) near the center and steeper ($\sim 1/r^3$) near the edge of the halo. Even though the NFW profile decreases more rapidly than Eq. (24.51) with increasing r , it can be shown that the total mass contained within the

NFW profile is still not bound (see Problem 24.23). In reality, other galaxies exist in our universe besides the Milky Way, and their mass density functions may overlap our own. As a result, although galaxies appear to be separate luminous objects, their dark matter halos may actually merge in intergalactic space.

A Component Model of the Milky Way Galaxy

Based on the mass density functions from star counts and kinematics, astronomers have been able to construct approximate models of the overall rotation curve of the Galactic disk. One such model is shown in Fig. 24.28. Note that the observational data show that the rotation curve is decreasing slightly in the solar neighborhood (R_0); hence the negative value for $d\Theta/dR = -(A + B)$ near the Sun. Although the model does not reproduce all of the fine structure that is present in the velocity data (which may be due to local density variations in the thin disk, such as spiral arms), the overall correspondence between the model and observations is quite good. Notice in particular the rigid-body rotation near the center, the local maximum due to the combined effects of the central bulge, the stellar halo and the dark matter halo, and the eventual flat rotation at large values of R .

Some Methods for Determining Distances

Before leaving the topic of Galactic kinematics, it is appropriate to discuss motion-based methods of determining distances within the Milky Way, the most important of which is the **moving cluster method**. Since in a stellar cluster the stars are gravitationally bound to one another, they move through space collectively. By recording the changing positions of members of the group over time due to the cluster's bulk motion, it is possible to determine their directions of motion. After removing the effects of the Sun's peculiar motion, tracing

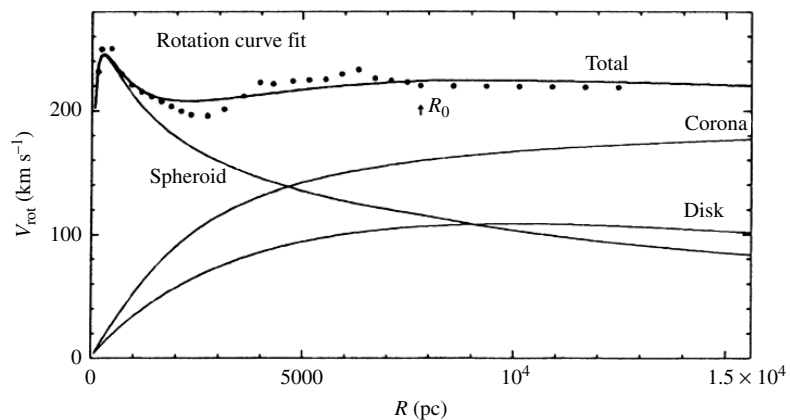


FIGURE 24.28 One model of the rotation curve of the Galaxy. The mass distribution of each Galactic component contributes to the overall velocity structure of the disk. The dots represent observational data. Note that the “spheroid” represents the bulge and stellar halo combined and the “corona” represents the dark matter halo. (Figure adapted from Gilmore, King, and van der Kruit, *The Milky Way as a Galaxy*, University Science Books, Mill Valley, CA, 1990.)

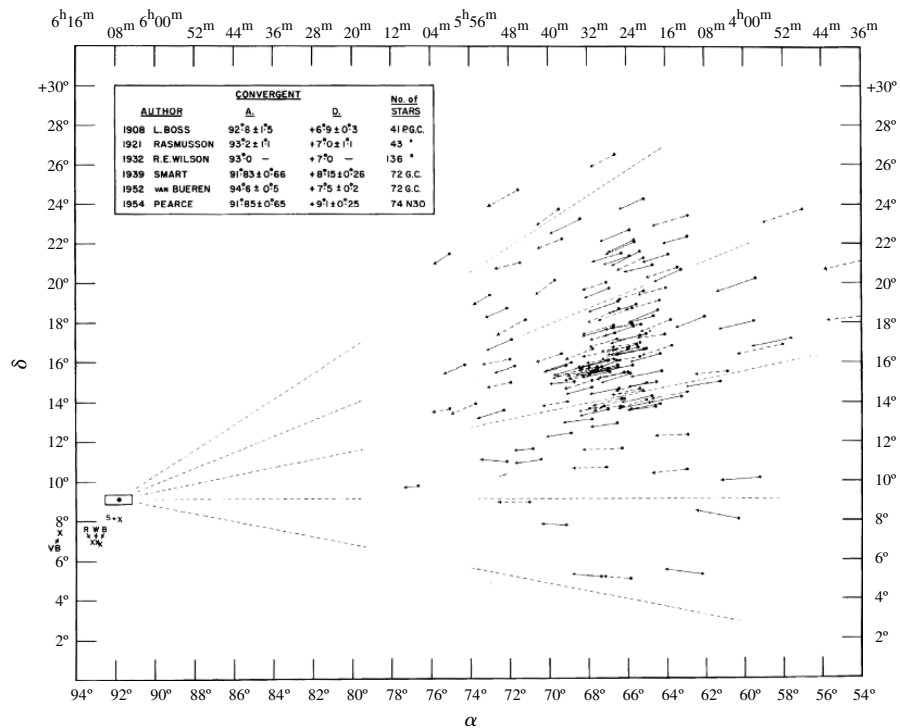


FIGURE 24.29 The apparent motion of the Hyades across the celestial sphere. (Figure adapted from *Elementary Astronomy* by Otto Struve, Beverly Lynds, and Helen Pillans. Copyright © 1959 by Oxford University Press, Inc. Renewed 1987 by Beverly T. Lynds. Reprinted by permission of the publisher.)

these directional vectors through space gives the appearance that each star is moving toward (or coming from) a common point known as the **convergent point**. This is simply the illusion that parallel lines appear to converge at infinity, a phenomenon familiar to anyone who has looked down the length of a long, straight set of railroad tracks. A diagram of the motion of the Hyades galactic cluster, found in the constellation of Taurus, is shown in Fig. 24.29.

From the geometry illustrated in Fig. 24.30, the angle between the cluster and the convergent point, as seen from the Sun, must be the same as the angle between the line of sight to the cluster and its space velocity vector, $\mathbf{v}$ (this statement is valid only because the convergent point is taken to be at infinity). Now, decomposing the space velocity into its perpendicular components, the radial velocity is given by $v_r = v \cos \phi$, and the transverse velocity is $v_t = v \sin \phi$. Combining yields

$$v_t = v_r \tan \phi.$$

Since the transverse velocity is observed as the proper motion, $\mu = v_t/d$, the distance to the cluster can be determined from knowledge of the direction to the convergent point ϕ ,

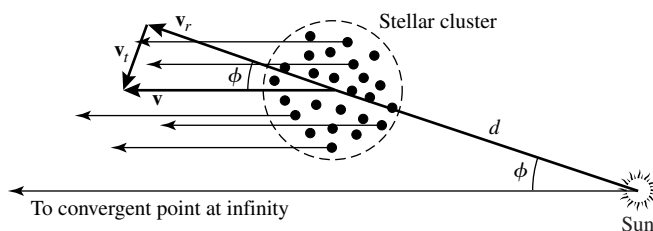


FIGURE 24.30 The space motion of the cluster is directed toward the convergent point. This velocity vector may be decoupled into its radial and transverse components.

the average radial velocity of the cluster members, and the average of their proper motions. Solving for d , we get

$$d = \frac{\langle v_r \rangle \tan \phi}{\langle \mu \rangle}, \quad (24.53)$$

where standard SI units have been used for d , v_r , and μ . Expressing d , v_r , and μ in the more commonly used units of pc, km s⁻¹, and arcsec yr⁻¹, respectively, Eq. (24.53) can be written as

$$d \text{ (pc)} = \frac{\langle v_r \rangle \tan \phi}{4.74 \langle \mu'' \rangle}. \quad (24.54)$$

This technique has been used to determine the distances to several clusters, including the Hyades with its approximately 200 members, the Ursa Major group (60 stars), and the Scorpio–Centaurus group (100 stars); the most important of these is the Hyades. The distance to the Hyades is found to be 46 ± 2 pc, in excellent agreement with other determinations such as those made by trigonometric parallax (the Hipparcos space astrometry mission data yielded a value of 47 pc to the center of the Hyades).

Once the distance to the Hyades was determined, the absolute magnitudes of its members could be found, providing an important calibration of its main sequence. By comparing the apparent magnitudes of other cluster H–R diagram main sequences to the Hyades, it is possible to find the distance moduli of those clusters, as illustrated schematically in Fig. 24.31. Assuming that the amount of interstellar extinction is known (from reddening data, for instance), the distances to those clusters can be determined. This distance technique, first discussed in Section 13.3, is known as **main-sequence fitting** and is similar to the method of spectroscopic parallax discussed at the end of Section 8.2. However, main-sequence fitting is a more precise procedure because it relies on a large number of stars along the main sequence rather than on a single object, significantly reducing statistical errors. Identifying RR Lyraes in clusters of known distances then provides a means of determining the intrinsic luminosities of these stars more accurately. Once the luminosities of the RR Lyraes have been calibrated, they can be used to determine other distances, such as to globular clusters. Historically, the Hyades provided the foundation for virtually all distance estimates, both Galactic and extragalactic, beyond about 100 pc from Earth, although this is changing with

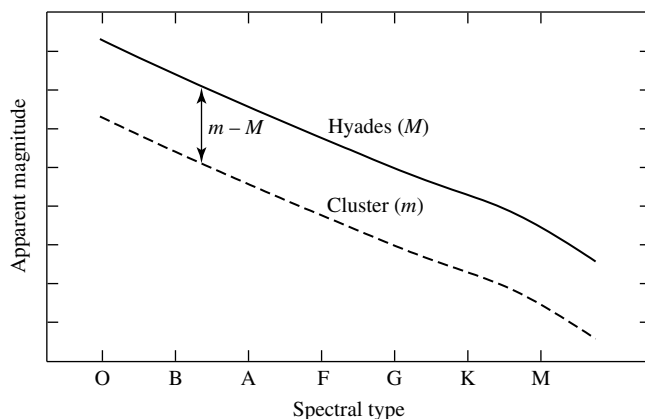


FIGURE 24.31 The distance modulus of a cluster can be determined by shifting the cluster's main sequence vertically in the H–R diagram until it coincides with the known absolute magnitude of the Hyades' main sequence.

space astrometric missions such as the European Space Agency's Hipparcos (1989–1993) and Gaia (launch 2011) missions, and NASA's SIM PlanetQuest (launch 2011).

One last distance determination technique worth mentioning is **secular parallax**. Recall from the discussion of trigonometric parallax in Section 3.1 that the ability to measure distances directly from the motion of Earth around the Sun relies on the 2-AU diameter of Earth's orbit. If the length of the baseline could be increased, it would be possible to extend the method to objects farther away. This is accomplished by using the peculiar motion of the Sun with respect to a group of stars having similar properties, such as similar spectral types, distances, and space motions. The overall solar motion of 13.4 km s^{-1} is equivalent to 2.8 AU yr^{-1} ; allowing the effect to build up for several years provides a significantly larger baseline than is available from Earth's annual orbit about the Sun. This larger baseline can then be used to determine an average distance to the group of stars. However, it is important to remember that the measured solar motion does depend on the group of stars used as a reference. We will not discuss the details of the method in this text.

24.4 ■ THE GALACTIC CENTER

Observations of the center of our Galaxy pose a particular challenge. This is because the abundance of gas and dust in the Galactic plane results in more than 30 magnitudes of extinction at visible wavelengths. Located only 30 pc above the midplane and 8 kpc from the center, the line of sight from the Sun to the center traverses nearly the maximum possible amount of interstellar material. It is interesting to note, however, that because the solar peculiar velocity has an appreciable component perpendicular to the Galactic plane ($w_{\odot} = 7.2 \text{ km s}^{-1}$), the Sun will reach a height of approximately 85 pc above the plane in 15 million years (see Problem 24.27); this will place it above most of the obscuring material.

If humanity is still inhabiting Earth at that time, our descendants will enjoy a spectacular view of a dense stellar cluster near the center of our Galaxy.

The Mass Distribution Near the Galactic Center

To peer into the central regions of the Galaxy today, we are forced to make most of our observations in wavelengths longer than about $1\ \mu\text{m}$ (i.e., infrared, microwave, and radio), or in X-rays and gamma rays. Despite our difficulty of clearly seeing the center of the Galaxy in visible wavelengths, we have been able to construct an image of the **Galactic nucleus** that depicts a history of violent events and exotic phenomena.

One IR wavelength band commonly used for investigations of the core of our Galaxy is centered at $2.2\ \mu\text{m}$, the so-called *K band*. This wavelength band is employed because the large number of old Population I K and M giant stars ($T_e \sim 4000\ \text{K}$) that exist in the central region of the Galaxy are readily observable at $2.2\ \mu\text{m}$. When we use the *K* band to study the brightness distribution of the central cluster and use an appropriate mass-to-light ratio ($\sim 1\ M_\odot/L_\odot$), it appears that the mass density of stars rises toward the center as $r^{-1.8}$ down to a radius of between $0.1\ \text{pc}$ and $1\ \text{pc}$. This is roughly the type of distribution that would be expected on dynamical grounds (the central region of the Galaxy is interior to the region of “rigid-body” rotation discussed on page 915).

Since these stars are very close together, particularly when compared with the distance between stars in the solar neighborhood, close encounters are fairly frequent, occurring on average once every 10^6 years or so. Because of the gravitational perturbations produced by these close encounters, the constant exchange of mechanical energy between stars has generated a nearly *isothermal* velocity distribution; in other words, the stars in the sample have a velocity distribution just like the particles in an isothermal gas, meaning that the velocity distribution is approximately Maxwellian. In a truly isothermal *stellar gas*, the mass density distribution is r^{-2} , close to the $r^{-1.8}$ variation that is observed. Recall that this is also the spherical density distribution required for flat rotation curves [when all stars have the same orbital velocity and $M_r \propto r$; Eq. (24.49)].

The observed density distribution from the isothermal stellar gas is inconsistent with measurements of the velocities of stars within $2\ \text{pc}$ of the center, however. Kristen Sellgren, Martina T. McGinn, and their colleagues made one such set of observations in the late 1980s using the $2.3\text{-}\mu\text{m}$ molecular absorption band of CO found in the spectra of the cool K and M giants. They found that although the velocity distribution is fairly isothermal from several hundred parsecs down to a couple of parsecs from the center, velocities begin to increase significantly as the distance to the center continues to decrease. This suggests that either the stellar density must rise substantially faster than r^{-2} toward the center (at least as steeply as $r^{-2.7}$) or there must be a great deal of mass occupying a very small volume near the middle of the cluster.

More recently, Rainer Schödel, Reinhard Genzel, and their research group were able to follow the orbits of stars very close to the Galactic center. In particular, the star known as S2 has an orbital period of $15.2\ \text{yr}$, an orbital eccentricity of $e = 0.87$, and a perigalacticon distance of $1.8 \times 10^{13}\ \text{m} = 120\ \text{AU}$ (17 light-hours). This size is only a few times the semimajor axis of Pluto’s orbit! Figure 24.32 shows the orbit of S2 against the central stellar cluster of the Galaxy.

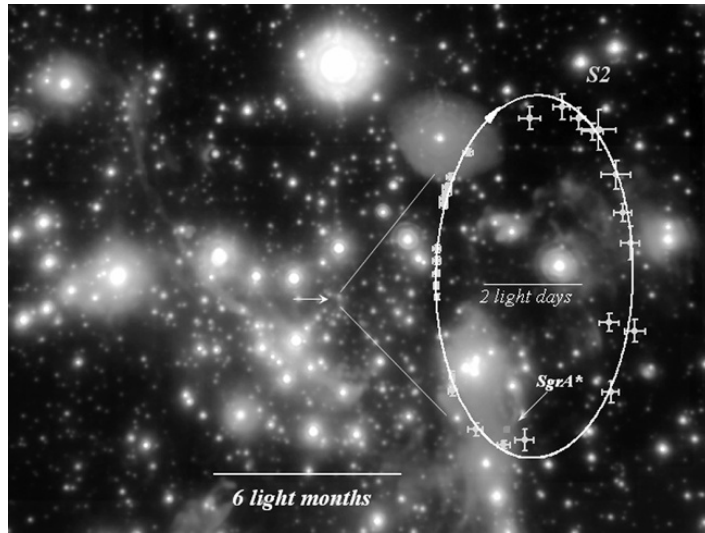


FIGURE 24.32 The orbit of S2 about the center of the Milky Way Galaxy. The center is designated as Sgr A*. (Courtesy of Reinhard Genzel and Rainer Schödel.)

Example 24.4.1. The semimajor axis of S2's orbit is (from Eq. 2.5)

$$a_{S2} = \frac{r_p}{1 - e} = 1.4 \times 10^{14} \text{ m.}$$

From Kepler's third law, Eq. (2.37), the mass interior to S2's orbit must be about

$$M = \frac{4\pi^2 a_{S2}^3}{G P^2} \simeq 7 \times 10^{36} \text{ kg} \simeq 3.5 \times 10^6 M_{\odot}.$$

A more precise calculation gives

$$M = 3.7 \pm 0.2 \times 10^6 M_{\odot}.$$

Figure 24.33 shows estimates of the amount of mass interior to r based on measurements of objects at varying distances from the center.

The luminosity distribution of the stars near the center of the Galaxy peaks within a few arcseconds (~ 0.1 pc) of an infrared object known as **IRS 16** (for infrared source). It was during a lunar occultation that IRS 16 was resolved into at least 15 very luminous point-like sources that are most likely individual stars. These sources appear to be hot stars with brightnesses in excess of $10^6 L_{\odot}$. They may be O and B stars, but with luminosities that are much greater than expected for normal stars. Furthermore, their ultraviolet fluxes are absorbed by the surrounding gas and dust and reradiated in the infrared. It has been suggested that these objects may be Wolf-Rayet stars, which are extremely rare in other parts of the Galaxy (recall Section 15.1). Since Wolf-Rayets are massive stars, they must have evolved to their present states very rapidly. If these stars are in fact Wolf-Rayets, then

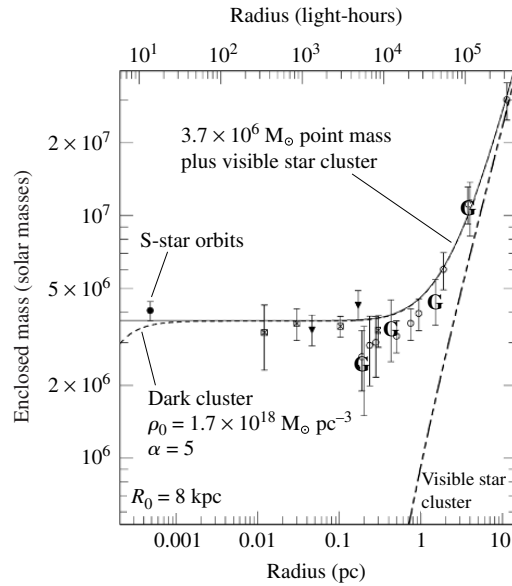


FIGURE 24.33 The interior mass function for the central 10 pc of the Galaxy. Note that the curve is consistent with a mass distribution $M_r \propto r$ beyond about 5 pc but that interior to 2 pc the distribution levels off, approaching a constant nonzero value of $3.7 \times 10^6 M_\odot$. “Dark cluster” refers to a hypothetical object. Note that the predictions of a dark cluster model at the center of the Galaxy do not agree with the observational data. (Adapted from a figure courtesy of Reinhard Genzel and Rainer Schödel. For a discussion of an earlier version of this diagram, see Schödel, et al., *Nature*, 419, 694, 2002.)

a burst of star formation must have occurred within the past 10 million years. However, an episode of recent star formation seems to contradict the facts that the density of gas and dust is very low and that there is no evidence of ongoing star formation in that region today. On the other hand, in further support of the Wolf–Rayet idea is the presence of high-velocity gas near IRS 16 ($\sim 700 \text{ km s}^{-1}$), which some researchers suggest may have been ejected as a stellar wind by one of the stars. As an alternative explanation, other astronomers point out that it is also possible that the high-velocity gas could simply be falling in toward the center of the Galaxy rather than being ejected from it.¹⁰ The velocity structure of the gas does not appear to be consistent with the acceleration expected of Wolf–Rayet winds with increasing distance from the star; instead, the gas velocity appears to decrease with distance.

IRS 16 does not appear to have sufficient mass to account for the rise in orbital velocities near the center. It seems that whatever the origin of this extremely localized mass distribution, it must have an overall luminosity that remains below our threshold of detectability. One possibility is that the mass is composed of a very dense cluster of brown dwarfs and/or more massive neutron stars. Even in the case of neutron stars, however,

¹⁰Determining whether the gas is falling in toward the center of the Galaxy or being ejected from it requires information about the orientation of the gas trajectories relative to the line of sight, something that is often difficult to determine.

interactions with high-mass stars would eject the lower-mass neutron stars from the center in less than 10^8 years. This would have the additional effect that the most massive members of the cluster would “sink” to the center of the Galaxy’s gravitational well, which is not observed.

Radio Sources in Sagittarius

It was Karl Jansky (1905–1950) who, in the 1930s, first discovered that a radio source was located in the direction of Sagittarius, but extensive observations of the radio Galaxy were delayed until after World War II.¹¹ Since that time the Galactic center has been the site of numerous studies at radio wavelengths.

Radio observations of H I clouds reveal a **nuclear disk** of neutral gas occupying a region between a few hundred parsecs and about 1 kpc from the center. The disk appears to be tilted slightly with respect to the Galactic plane and contains clouds with significant noncircular motions. Observations of the nuclear disk are responsible for the mass estimates of the Galactic center beyond 100 pc that are shown in Fig. 24.33.

Galactic lobes of ionized gas have also been observed within the central few hundred parsecs. Elongated, with dimensions of 10 pc by 200 pc, the lobes are oriented nearly perpendicular to the Galactic plane and may represent material being ejected from the center. However, the data may also be consistent with infalling material.

One of the more unusual features of the radio emission originating in the central region of the Galaxy is the set of filaments that stretch for 20 pc from the center in a direction perpendicular to the Galactic plane and then make an almost right-angle turn; see Fig. 24.34. Even a casual inspection of this 20-cm radio structure, located near $\ell \sim 0.18^\circ$, suggests that magnetic fields may be responsible for the unusual pattern. In fact, the radiation is linearly

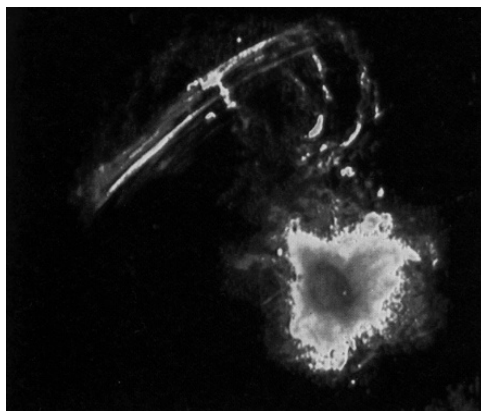


FIGURE 24.34 A view of the central $60 \text{ pc} \times 60 \text{ pc}$ of the Galaxy. This image was made using 20-cm radiation produced by synchrotron radiation. Sgr A is the central, radio-bright region. (Figure from Yusef-Zadeh, Morris, and Chance, *Nature*, 310, 557, 1984, and NRAO.)

¹¹Major advances in radio and microwave electronics that occurred during World War II helped to advance radio astronomy in the years that followed.

polarized and appears to be synchrotron emission. From the intensity of the radiation and the amount of polarization, the strength of the Galactic magnetic field in that region is probably between 10^{-8} and 10^{-6} T, roughly two to four orders of magnitude weaker than Earth's magnetic field.

A similar feature has also been discovered south of the Galactic plane. Together, these structures seem to be part of a “halo” of streamers and filaments that may correspond to mass outflow from the center of the Galaxy. A satisfactory model of the source of the relativistic electrons that are spiraling around the magnetic field lines, or of the source of the field itself, has not yet been developed.

The inner 8 pc of the Galaxy contains the radio source designated as **Sagittarius A** (Sgr A). With the increase in resolving power provided by the technique of very long baseline interferometry and the use of telescopes such as the Very Large Array, it has become possible to study the structure of the Sgr A complex down to angular resolutions of 0.2 milliarcsecond, corresponding to linear dimensions of less than 2 AU.

The largest feature of the complex is the **molecular circumnuclear ring**, a doughnut-shaped structure with an inner radius of 2 pc and an outer radius of 8 pc that is inclined some 20° with respect to the Galactic plane. The inner edge of the ring exhibits some warping, and the ring thickness increases from 0.5 pc at its inner edge to 2 pc at a Galactocentric distance of 7 pc. This molecular ring has been observed at various wavelengths associated with several atoms and molecules, including H I, H_2 , C II, O I, OH, CO, HCN, and CS. The ring is rotating about the Galactic center at a speed of approximately 110 km s^{-1} , a value that is nearly independent of radius. From the collisional excitation of the molecules and the intensity of the emission, the estimated mass of the portion of the ring located between 2 and 5 pc is $1 \times 10^4 M_\odot$ to $3 \times 10^4 M_\odot$.

It is apparent that the ring is unlike any other molecular region known to exist in our Galaxy. For instance, the temperatures of individual molecular clouds that exist throughout much of the ring increase from less than 300 K near 4.5 pc to more than 400 K near the inner edge. At the same time, the number density of hydrogen molecules increases from $1.5 \times 10^{10} \text{ m}^{-3}$ to $5 \times 10^{10} \text{ m}^{-3}$ over the same distance. These values should be compared with those of more typical molecular clouds, such as the giant molecular clouds discussed in Section 12.1. In GMCs that are located several kiloparsecs from the center of the Galaxy, $T \sim 15 \text{ K}$ and $n_{\text{H}_2} \sim 10^8 \text{ m}^{-3}$.

The circumnuclear ring also shows evidence that some violent event occurred near the Galactic center in the relatively recent past. The inner edge of the ring is very sharp; the number density of particles inside the central cavity is 10 to 100 times less than in the ring itself. Such a strong density discontinuity cannot be an equilibrium feature of the ring because the ring's internal turbulence would destroy the discontinuity in less than 10^5 years. Furthermore, the gas within the cavity is largely ionized while the gas in the ring is in the form of neutral atoms and molecules. It is estimated that the amount of energy required to clear out the cavity is on the order of 10^{44} J , a value characteristic of a supernova explosion.

The ring shows other evidence of some past violent episode as well. For instance, material in the ring is very clumpy, a situation that cannot be maintained indefinitely because of the relatively rapid smoothing effects of cloud–cloud collisions. Also, a study of the hydroxyl molecule (OH) in portions of the ring indicates that the temperature of nearly 2000 K is consistent with strong shocks that rapidly heated the molecules to temperatures well above

the values seen elsewhere in the ring. Furthermore, OH radicals can be produced when water molecules (H_2O) are torn apart by energetic events such as shocks.

Besides the circumnuclear ring, several other components have also been discovered within the Sgr A complex. **Sgr A East** is a nonthermal source with a shell-like appearance (a “nonthermal” source is one that *does not* emit its electromagnetic radiation in the form of a blackbody spectrum; one type of nonthermal emission is synchrotron radiation). It is generally accepted that Sgr A East is a young supernova remnant that may be between 100 and 5000 years old. **Sgr A West**, which is located $1.5'$ from the center of Sgr A East, is an unusual H II region (a thermal source) that looks very much like a “mini-spiral.” Finally, **Sgr A*** (pronounced “Sagittarius A star”) is a strong, unresolved radio point source that is located near the center of Sgr A West. A radio image of Sgr A West and Sgr A* is shown in Fig. 24.35(a).

The location of the Sgr A complex corresponds very closely with the brightness peak of the central stellar cluster, suggesting strongly that this region marks the center of the Galaxy. In fact, one component of IRS 16 (named IRS 16 Center) is only about $1''$ west of Sgr A*. If the two objects are the same distance from Earth and not simply aligned along the line of sight, then their angular separation translates into a linear separation of only 0.04 pc. For reasons we will soon discuss more fully, it now appears that Sgr A* is the actual center of the Milky Way.

Although Sgr A West appears superficially to be much like the large-scale spiral pattern seen in the disk of the Milky Way, it is fundamentally very different. The western arc is just the ionized inner edge of the circumnuclear ring, which is rotating about the center at

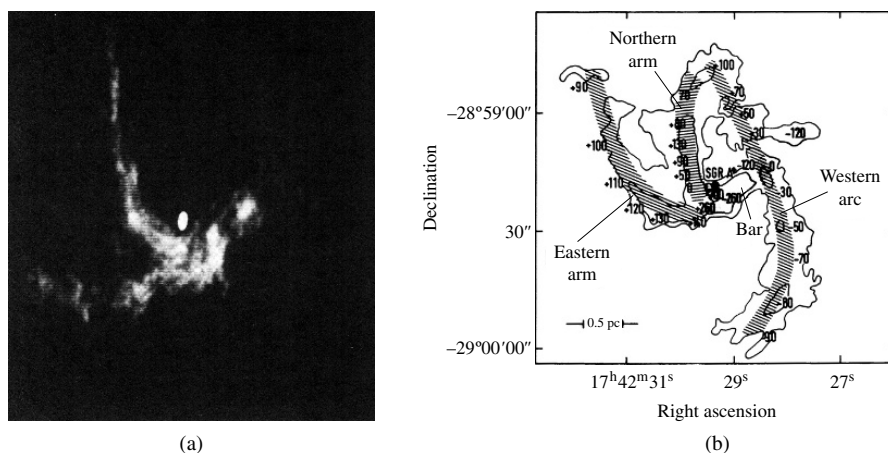


FIGURE 24.35 (a) A VLA 6-cm image of the inner $40''$ of the Galaxy, showing the Sgr A West mini-spiral. Sgr A* (a radio point source) is the bright oval at the center of the image. (Figure from Lo, *Science*, 233, 1394, 1986.) (b) An intensity contour map of the Sgr A West region. Radial-velocity measurements of the gas are also indicated on the map (in units of km s^{-1}). (Figure from Genzel and Townes, *Annu. Rev. Astron. Astrophys.*, 25, 377, 1987. Reproduced with permission from the *Annual Review of Astronomy and Astrophysics*, Volume 25, ©1987 by Annual Reviews Inc.)

110 km s⁻¹. The velocities recorded on the map in Fig. 24.35(b) are the measured radial velocities. Portions of the arc that are moving nearly along the line of sight record the largest radial velocities (positive or negative), and the parts of the arc that are moving perpendicular to the line of sight have no radial-velocity components.

The other components of Sgr A West appear to be ionized filaments of matter that may be both rotating about and falling in toward the central “bar” and the vicinity near Sgr A*. These features may also be associated with a central magnetic field that has an estimated strength of 30 nT.

As was seen for the stellar velocities, the velocities of the ionized gas within the cavity increase dramatically toward the center, from 110 km s⁻¹ at the inner edge of the circum-nebular ring to 700 km s⁻¹ at 0.1 pc.

Example 24.4.2. The motion of gas in the central regions of the Galaxy can be used to estimate the amount of mass interior to the location of the gas. A gas cloud 0.3 pc from the center has a measured velocity of 260 km s⁻¹. If the cloud is in orbit about the center, Eq. (24.49) gives

$$M_r = \frac{v^2 r}{G} = 4.7 \times 10^6 M_\odot.$$

An X-Ray Source in Sgr A

To within the positional error bars, the radio Sgr A West region (including Sgr A*) also coincides with a small continuous X-ray source. Although the estimates are highly uncertain due to extensive absorption along the line of sight, the X-ray source appears to have a characteristic temperature of $T \sim 10^8$ K and luminosities of 10^{28} W in the 2–6 keV energy band (“soft” X-rays) and 2×10^{31} W between 10 keV and 10 MeV (“hard” X-rays). Because the X-ray source is also highly variable, it must be composed of one or a few objects whose diameters are less than about 0.1 pc. The upper limit on the linear size is derived from the minimum amount of time required for information to travel across the object, assuming that the information travels at the speed of light. If one side of the source begins changing brightness, the opposite side can learn about the change and begin contributing to it only after a time $\Delta t \geq d/c$, where d is the diameter; measuring Δt places the upper limit of 0.1 pc on d .

The Supermassive Black Hole in Sgr A*

As we saw in Example 24.4.1, it has been possible to follow the orbits of individual stars to within about 120 AU of Sgr A*, providing us with critical data for accurately calculating the mass in the innermost region of the Galactic center. It has also been possible to place an upper limit of less than 2 AU on the size of Sgr A*. From these data, it appears that Sgr A* can only be a **supermassive black hole** with a mass of

$$M_{\text{Sgr A}^*} = 3.7 \pm 0.2 \times 10^6 M_\odot.$$

This implies that the Schwarzschild radius of the black hole is (Eq. 17.27)

$$R_{\text{Sgr A}^*} = \frac{2GM_{\text{bh}}}{c^2} = 0.08 \text{ AU} = 16 R_{\odot},$$

which remains below the roughly 2-AU resolution limit of current observations.

Producing the infrared radiation detected from the ring and maintaining the degree of ionization seen inside the cavity require an ultraviolet luminosity of $10^7 L_{\odot}$ with an effective blackbody temperature of 35,000 K; the temperature is deduced from the strengths of the IR emission lines of S IV, O III, and Ar III. The gas absorbs some of the UV radiation, ionizing it and producing the unusual H II region, Sgr A West. The dust absorbs the remainder of the UV photons and reradiates the light at IR wavelengths.

Could a supermassive black hole be responsible for the luminosity? Observations of the number density of particles and the velocity structure inside the cavity of the circumnuclear ring suggest that matter is accreting onto the Galactic center at a rate of $\dot{M} = 10^{-3}$ to $10^{-2} M_{\odot} \text{ yr}^{-1}$. This matter must release gravitational potential energy as its distance from the center decreases.

It is possible to make a rough estimate of the luminosity that could be produced by an accreting supermassive black hole in Sgr A*. If we consider a Newtonian view of energy released when a particle of mass M spirals in through an accretion disk from an initial radius of r_i to a final radius of r_f , then according to the virial theorem, the amount of energy radiated should be one-half the change in potential energy, or

$$E = \frac{1}{2} \left(\frac{GM_{\text{bh}}M}{r_f} - \frac{GM_{\text{bh}}M}{r_i} \right),$$

where M_{bh} is the mass of the black hole. Assuming that $r_i \gg r_f$ and $r_f = R_S$ (the Schwarzschild radius), then

$$E = \frac{1}{2} \frac{GM_{\text{bh}}M}{R_S}.$$

Taking the luminosity as $L = dE/dt$ and the mass accretion rate as $\dot{M} = dM/dt$, and substituting the expression for the Schwarzschild radius, we have

$$L = \frac{1}{4} \dot{M} c^2, \quad (24.55)$$

a result that is independent of both the mass and the radius of the black hole. Now, the minimum mass-accretion rate required to generate $10^7 L_{\odot}$ is

$$\dot{M} = \frac{4L}{c^2} = 1.7 \times 10^{17} \text{ kg s}^{-1} = 2.7 \times 10^{-6} M_{\odot} \text{ yr}^{-1}.$$

The observed accretion rate of 10^{-3} to $10^{-2} M_{\odot} \text{ yr}^{-1}$ is more than sufficient to produce the luminosity seen in Sgr A West and Sgr A*. Curiously, high-resolution observations of the Sgr A* region alone suggests an upper limit of luminosity from Sgr A* of less than $3 \times 10^4 L_{\odot}$.